Separable Rank of Radial Kernels in Two and Higher Dimensions: Machine-Checked Proofs in Lean 4 and Isabelle/HOL

GitHub Copilot

Source archive (CC0 1.0): https://github.com/OlliNiemitalo/kernel-rank-lean

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Abstract

A d-variate kernel $K : X_0 \times \cdots \times X_{d-1} \to \mathbb{K}$ has *separable rank* at most N if it is a sum of N coordinatewise products $\sum_{n=0}^{N-1} \prod_{j=0}^{d-1} f_{n,j}(x_j)$. On a d-dimensional Cartesian grid of support width L, a rank-R kernel evaluates in dRL one-dimensional taps per grid point rather than L^d. While a positive Gaussian $\exp(-\alpha\|x\|^2)$ has separable rank 1, approximating flat-top or oscillatory isotropic kernels—such as defocus disks, Airy point-spread functions, circular low-pass filters, or band-limited wave propagators—with finite coordinatewise separable rank relies on signed or oscillatory profiles built from complex-rate Gaussians and exponential polynomials.

We give a complete machine-checked characterization in Lean 4 (with Mathlib) of the d-way separable rank of continuous circularly and spherically symmetric kernels $K(x) = \varphi(\|x\|^2)$ on $\mathbb{R}^d$ for all $d \geq 2$: (1) $\operatorname{separableRank}(K) < \infty$ over $\mathbb{R}$ if and only if φ agrees on $[0, \infty)$ with a well-formed exponential-polynomial representation $\sum_i \operatorname{Re}(P_i(s)e^{\lambda_i s})$ (nonzero polynomials P_i, distinct rates λ_i with $\operatorname{Im}\lambda_i \geq 0$, and $P_i \in \mathbb{R}[X]$ when $\operatorname{Im}\lambda_i = 0$); (2) over $\mathbb{C}$, a continuous radial profile $\psi(\|x\|^2)$ has exact d-way separable rank $\sum_{\mu \in S}((d-1)\deg P_\mu + 1)$ for nonzero polynomials P_μ at distinct rates; (3) over $\mathbb{R}$, a continuous radial profile $\varphi(\|x\|^2)$ has exact d-way separable rank $\sum_i c_d(P_i, \lambda_i)$ where each real-rate component ($\operatorname{Im}\lambda_i = 0$) costs $(d-1)\deg P_i + 1$ real d-way products (achieved by real Lagrange interpolation) and each nonreal conjugate-pair component ($\operatorname{Im}\lambda_i > 0$) costs $R_i = 2(d-1)\deg P_i + d$ real d-way products, achieved via an alternating R_i-point root-of-unity polynomial-product identity (giving d rather than 2^{d-1} real products for a complex Gaussian when $\deg P_i = 0$) and proved optimal by Alder–Strassen degree peeling over $\mathbb{R}$; and (4) in $d = 2$, scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ preserves rank unconditionally and exact rank equals the degree of the minimal monic constant-coefficient half-line differential annihilator $p(D)\varphi_{\mathbb{C}} = 0$, whereas for every $d \geq 3$ scalar extension and compact-open limits can strictly lower rank: we prove in Lean 4 that $\|x\|^2 \cos(\|x\|^2)$ on $\mathbb{R}^d$ has exact complex rank $2d$, exact real rank $3d - 2$, and is the uniform limit on every compact set of real kernels of rank at most $2d$ (giving an exact-minus-border real rank gap of at least $d - 2$).

Necessity reduces to showing that a continuous one-parameter semigroup $M : [0, \infty) \to \mathcal{A}$ is an exponential $M(t) = \exp(tA)$. Comparing the 96-line Banach-algebra integral-smoothing proof in Lean 4 against a 256-line finite-difference proof and three finite-dimensional nonstandard proofs (206 and 399 lines in Lean 4, 275 lines in Isabelle/HOL) shows that library-amortized integration remains 2×–4× shorter and strictly more general even after a homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$ eliminates the 124-line *a priori* difference-quotient bound. Finally, we give a visual walkthrough of 1-D/2-D kernel formation and streaming intermediates, connect the root-of-unity identity to planar harmonic ("beach-ball") differential operators, and evaluate two GPU-verified numerical applications on an NVIDIA RTX 4070 Ti Super: (i) numerically optimized Chebyshev 2-D circular lens blur, where nuclear-norm-optimal $\mathrm{GL}_N(\mathbb{R})$ rebalancing on $\operatorname{rowSpan}(K)$ eliminates up to 1,110× intermediate cancellation so `float16` buffers stay

within 0.17/255 of `float64` (unlike $d \geq 3$, where minimal CP decompositions lack transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotation freedom), and (ii) explicit 3-D acoustic wave-equation leapfrog propagators, identifying Kowalczyk and van Walstijn's 27-point IWB stencil as the single-real-Gaussian ($R = 1$) baseline and designing von Neumann-stable multi-Gaussian propagators ($R = 4$–9) that cut directional phase-velocity error up to 0.80π from 9.38%–19.88% to 0.049% (an analytical resolution-scaling speedup of 476×–660× and 865× fewer grid voxels than 27-point IWB at a 0.1% dispersion tolerance).

1 Introduction

Convolving a d-dimensional array with a dense $L \times \cdots \times L$ spatial kernel requires L^d multiply–accumulate operations per output sample. When the kernel $K : \mathbb{R}^d \to \mathbb{K}$ admits a rank-R coordinatewise decomposition

$$K(x_0, \ldots, x_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} f_{n,j}(x_j), \tag{1}$$

the d-dimensional convolution reduces to R cascades of d one-dimensional convolutions, costing dRL operations per sample (or $O(dR)$ operations independent of L when the one-dimensional factors admit recursive causal/anti-causal realizations). The least $R \in \mathbb{N} \cup \{\infty\}$ for which (1) holds is the *separable rank* of K: matrix rank when $d = 2$, and functional canonical polyadic (CP) tensor rank when $d \geq 3$.

For circularly or spherically symmetric (*radial*) kernels in dimension $d \geq 2$,

$$K(x) = \varphi\left(\sum_{j=0}^{d-1} x_j^2\right), \qquad x = (x_0, \ldots, x_{d-1}) \in \mathbb{R}^d,$$

Maxwell's characterization of the normal distribution implies that a continuous nonvanishing radial kernel of separable rank 1 has a Gaussian profile $\varphi(s) = \alpha e^{\lambda s}$. Real-rate Gaussians ($\lambda < 0$), however, are strictly monotone and convex in $s = \|x\|^2$; nonnegative combinations of them cannot exhibit flat passbands, sharp cutoffs, or zero crossings. In image processing, optics, and numerical wave propagation, O. Niemitalo [17, 20, 21] and related filter-design work [13] approximate non-Gaussian radial profiles—such as circular defocus disks, Airy diffraction patterns $(J_1(r)/r)^2$, and ideal circular low-pass filters $\omega_c J_1(\omega_c r)/(2\pi r)$—by sums of *complex-rate* Gaussians $\mathrm{Re}(\beta e^{(-\alpha+i\omega)\|x\|^2})$, whose damped radial oscillations $\cos(\omega\|x\|^2 + \theta)$ synthesize sharp transitions and sidelobe rings while keeping finite separable rank.

1.1 Visual overview: components, rendering terms, and streaming intermediates

Before introducing algebraic and functional-analytic machinery, Figure 1 illustrates concretely how one-dimensional passes synthesize a circularly symmetric 2-D filter and why the distinction between *components* and *rendering terms* matters in practice:

1. **Mathematical "Components" view (Figure 1a):** Although a real bell curve $e^{-b(x^2+y^2)} = e^{-bx^2}e^{-by^2}$ cannot form a flat-top circle, allowing a *complex* rate $\lambda = -b + ic$ preserves the 1-D split $e^{\lambda(x^2+y^2)} = e^{\lambda x^2}e^{\lambda y^2}$ while oscillating radially as $e^{-br^2}\cos(cr^2)$ and $e^{-br^2}\sin(cr^2)$ ($r^2 = x^2 + y^2$). Writing $g(x) = \sqrt{a}\, e^{\lambda x^2} = g_{\mathrm{re}}(x) + i g_{\mathrm{im}}(x)$, taking the real part of $g(x)g(y)$ requires two real 1-D×1-D products:

$$\mathrm{Re}\big(g(x)g(y)\big) = g_{\mathrm{re}}(x)\, g_{\mathrm{re}}(y) + g_{\mathrm{im}}(x)\, \big(-g_{\mathrm{im}}(y)\big).$$

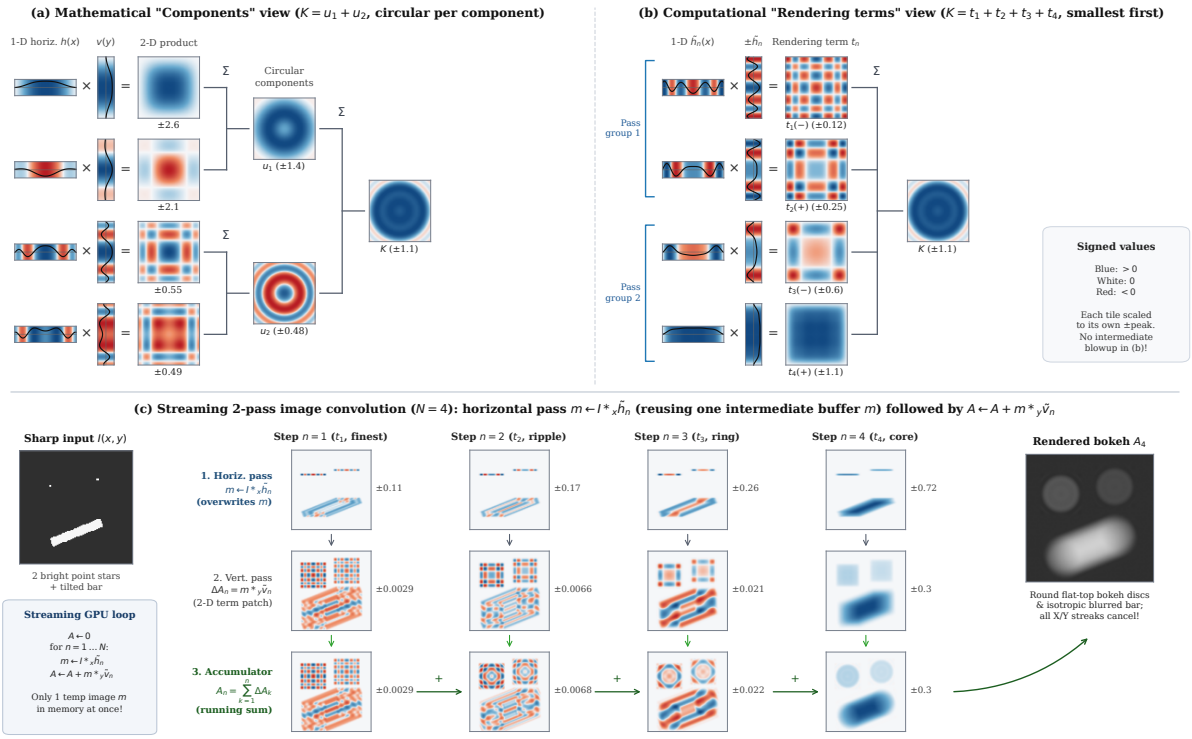


Figure 1: Visual guide to separable radial filtering on a two-complex-Gaussian (2c, exact rank $N = 4$) circular disc-blur kernel [21]: (a) mathematical *Components* view, where pairs of 1-D products $h(x)v(y)$ form circularly symmetric concentric-ring components u_1, u_2 that sum to the flat-top disc K; (b) computational *Rendering terms* view, where orthogonalizing rowSpan(K) yields four direct real products $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered smallest-first so peak amplitudes never blow up; (c) streaming two-pass image blur using *one* overwritten horizontal intermediate buffer $m \leftarrow I *_x \tilde{h}_n$ followed by vertical accumulation $A \leftarrow A + m *_y \tilde{v}_n$, canceling all axis-aligned streaks into round bokeh discs and an isotropically blurred bar.

Neither outer product is circular on its own—each is a rectangular plaid of positive (blue) and negative (red) stripes—yet adding the pair cancels every diagonal corner artifact to form a circularly symmetric component (u_1: wide dome; u_2: concentric sharpening rings), and $u_1 + u_2$ approximates a flat-top disc K. (More generally, when a component multiplies $e^{\lambda r^2}$ by a polynomial $P(r^2)$ of degree $m - 1 \geq 1$, expanding $(x^2 + y^2)^k$ naively across monomials would create $m(m + 1)$ real products, whereas collecting powers of x^2 merges them into $2m$ real products, or m when $c = 0$.)

2. **Computational "Rendering terms" view (Figure 1b):** A renderer never needs the intermediate circular components u_1, u_2—only the final sum $K(x, y) = \sum_{n=1}^{N} \tilde{h}_n(x)\tilde{v}_n(y)$. In high-accuracy filters (such as 6 complex Gaussians, $N = 12$), individual components u_i grow hundreds of times larger than K and cancel one another, causing severe rounding loss in 16-bit half-precision (`float16`) GPU buffers. Orthogonalizing the 1-D filter subspace (`separableChannels` in [21]; Section 5.2) rotates the N filter pairs into orthogonal *rendering terms* $t_n(x, y) = \tilde{h}_n(x) \cdot (\pm\tilde{h}_n(y))$ ordered from smallest to largest magnitude: every term stays bounded at or below the kernel's own peak (± 0.12 to ± 1.1 in Figure 1b, vs. ± 2.6 in Figure 1a), and brackets mark pairs of terms that can share a single complex pass.

3. **Streaming two-pass convolution with one intermediate buffer (Figure 1c):** To blur an input image $I(x, y)$, the GPU initializes an output accumulator $A \leftarrow 0$ and loops over $n = 1, \ldots, N$ (smallest term first, so fine corrections are not swamped in low precision).

In step n, a 1-D horizontal pass writes $m(x,y) \leftarrow (I *_x \tilde{h}_n)(x,y)$, smearing bright point stars horizontally into 1-D signed streaks; crucially, *the same temporary buffer m is overwritten on every step*, keeping working memory constant at one intermediate image regardless of N. A 1-D vertical pass then computes $\Delta A_n = m *_y \tilde{v}_n$ and adds it into A: steps $n = 1, 2, 3$ build a crisp sharpening ring A_3, and step $n = 4$ fills in the soft core so all horizontal and vertical streaks vanish identically in A_4.

This concrete picture motivates three foundational questions investigated in the rest of the paper:

1. **Classification ($d \geq 2$):** Are damped/oscillatory exponential-polynomial profiles in $s = \|x\|^2$ *universal*—must every continuous radial kernel $K(x) = \varphi(\|x\|^2)$ of finite d-way separable rank have this exact form?
2. **Exact d-way rank over $\mathbb{C}$ and over $\mathbb{R}$:** What is the exact minimum number of d-way rendering terms in (1) for any such representation? In $d \geq 3$ dimensions (such as 3-D acoustic simulation), can algebraic root-of-unity combinations beat naive real-and-imaginary expansion, and how does loss of 2-D's basis-rotation freedom affect conditioning and border rank?
3. **Proof architecture in interactive theorem provers:** In formalizing necessity—whose analytic core is the continuous one-parameter semigroup theorem $M(t) = \exp(tA)$ on $[0,\infty)$—how do finite-difference and nonstandard infinitesimal proofs compare in length, library reliance, and generality against Banach-algebra integral smoothing in Lean 4 and Isabelle/HOL? (As Section 4 shows, integral smoothing remains 2×–4× shorter and holds in every Banach algebra even after homogeneous coordinates remove the 124-line difference-quotient bound.)

1.2 Contributions and overview

All numbered theorems, propositions, and lemmas in Sections 2–4 (20 top-level Lean theorems and 1 Isabelle theorem audited by `scripts/check`) are machine-checked without unproved steps (`sorry` in Lean 4, oracles in Isabelle/HOL) in the open-source repository `kernel-rank-lean`, using Lean `v4.36.0-rc2` with Mathlib `v4.36.0-rc2` and Isabelle2026; Section 5 develops analytical and GPU-verified numerical applications:

- **Unified d-dimensional classification (Theorem 3.3, `separableRank_radial_lt_top_iff_exists`):** For every $d \geq 2$ and every $\varphi : \mathbb{R} \to \mathbb{R}$ continuous on $[0,\infty)$, the radial kernel $x \mapsto \varphi(\sum_{j=0}^{d-1} x_j^2)$ has finite d-way separable rank if and only if φ agrees on $[0,\infty)$ with a canonical real exponential-polynomial representation $\sum_i \mathrm{Re}(P_i(s)e^{\lambda_i s})$.
- **Exact d-D complex radial rank (Theorem 3.1, `separableRank_radial_complex_eq_iff`):** For $d \geq 2$ and $\psi : \mathbb{R} \to \mathbb{C}$ continuous on $[0,\infty)$, $\mathrm{separableRank}(x \mapsto \psi(\|x\|^2)) = N$ if and only if $\psi(s) = \sum_{\mu \in S} P_\mu(s)e^{\mu s}$ on $[0,\infty)$ for distinct rates $\mu \in S$ and nonzero polynomials P_μ satisfying $\sum_{\mu \in S}((d-1)\deg P_\mu + 1) = N$. The upper bound uses shifted root-of-unity Fourier extraction on truncated exponential polynomials; the lower bound uses multi-mode Alder–Strassen degree peeling.
- **Exact d-D real radial rank and border-rank gap (Theorems 3.5 and 3.7):** For every $d \geq 2$, $\mathrm{separableRank}(x \mapsto \varphi(\|x\|^2)) = N$ over $\mathbb{R}$ (`separableRank_radial_eq_iff`) if and only if φ agrees on $[0,\infty)$ with a well-formed representation whose d-dimensional cost (`multiTotalCost`) equals N, counting $(d-1)\deg P_i + 1$ for real rates ($\mathrm{Im}\,\lambda_i = 0$, via real Lagrange interpolation) and $R_i = 2(d-1)\deg P_i + d$ for nonreal rates ($\mathrm{Im}\,\lambda_i > 0$, via the alternating root-of-unity polynomial-product identity `re_coeff_prod_eq_sum_re` and

Alder–Strassen peeling over $\mathbb{R}$). Specializing to $\deg P_i = 0$ gives $r + dc$ (d real terms rather than 2^{d-1} per complex Gaussian, `separableRank_radial_gaussian_profile`), and specializing to $d = 2$ recovers totalCost (`separableRank_radial_two_eq_iff`) and the minimal differential annihilator (Theorem 3.6). We also formalize in Lean 4 that $\|x\|^2 \cos(\|x\|^2)$ on $\mathbb{R}^d$ has exact real rank $3d - 2$, complex rank $2d$, and compact-open border rank at most $2d$ (Theorem 3.7).

- **Comparative formalization of the semigroup step (Section 4):** We compare the 96-line Banach-algebra integral-smoothing proof with a 256-line finite-difference proof in Lean 4 and three finite-dimensional nonstandard proofs (206 lines in Lean via ultrafilter sequences, 399 lines in Lean via an ultrapower interface, and 275 lines in Isabelle/HOL), all simplified by the homogeneous common-scale normalization $\eta = \varepsilon + \|M(\varepsilon) - 1\|$.
- **Steerable operators, 2-D lens blur, and 3-D acoustic propagation (Section 5):** Specializing the root-of-unity identity to equal directional-derivative factors recovers odd and even planar "beach-ball" harmonic operators $\mathrm{Re}((\partial_x + i\partial_y)^m)$ [19, 22] and pass-count bounds for steerable derivative banks [18]. Complementing the formal proofs, reproducible numerical scripts and GPU benchmarks in `Applications` evaluate (i) 2-D circular disc blur across 18 component families, showing how $\mathrm{GL}_N(\mathbb{R})$ rebalancing on $\mathrm{rowSpan}(K)$ removes 595×–1,110× channel cancellation for `float16` rendering, and (ii) explicit 3-D acoustic wave propagators optimized across 232,620 Brillouin-octant and spherical-shell wavevectors, reducing isotropic dispersion up to 0.80π by two orders of magnitude over 7-point and 27-point FDTD schemes [10].

1.3 Related work

Radial characterizations, Gaussian sums, and wave propagators. Maxwell's kinetic-theory argument (1860) and Schoenberg's characterization of completely monotone radial functions [26] identify single Gaussians and positive Gaussian mixtures among radial kernels. For oscillatory or compactly supported radial kernels in scientific computing, Beylkin and Monzón [3] approximate profiles and Green's functions by short sums of complex exponentials and Gaussians; in image filtering, Lampert and Wirjadi [13] study anisotropic Gaussian separations, while Niemitalo [17, 20, 21] fits circular defocus and low-pass profiles directly with complex-variance Gaussians. In numerical acoustics and seismic wave propagation, Kowalczyk and van Walstijn [10] and Savioja and Svensson [25] analyze 27-point compact explicit FDTD stencils to mitigate grid dispersion, while Fomel, Ying, and Song [6] approximate Fourier-domain wave-extrapolation symbols by low-rank separable expansions; Section 5.3 bridges these two approaches in physical space via certified von Neumann-stable radial multi-Gaussian stencils. None of the prior works establishes a machine-checked converse for arbitrary signed finite-rank radial kernels or exact d-way real CP ranks across higher-degree components.

Functional equations, Hankel operators, and real vs. complex tensor rank. Reducing radial kernels via $s_j = x_j^2$ turns K into the d-way additive Hankel kernel $\varphi(s_0 + \cdots + s_{d-1})$. In two variables, functional equations of the form $f(x + y) = \sum_{n=1}^{N} X_n(x) Y_n(y)$ go back to Levi-Civita [15], Stone [27], Anselone and Korevaar's theorem on finite-dimensional translation-invariant subspaces [2], and Laczkovich [12], alongside Kronecker's theorem (1881) on finite-rank discrete Hankel matrices; our continuous half-line formalization uses operator-norm continuous shift semigroups on $\mathcal{W}(\varphi)$ to avoid differentiability hypotheses beyond continuity. In $d \geq 3$ variables, symmetric and Hankel tensor decompositions go back to Sylvester's apolarity theory (1851) and Comon and Mourrain [4]; Kruskal [11] established generic essential uniqueness of minimal three-way CP decompositions, Landsberg [14] treats multilinear rank substitution (originating in Alder and Strassen [1]) and border rank, while Reznick [24] and Qi, Comon,

and Lim [23] show that real symmetric/CP rank can strictly exceed complex rank. Our Theorems 3.1 and 3.5 give exact closed-form complex and real d-way ranks for the entire continuous Hankel exponential-polynomial class, pinning down the exact real-over-complex gap ($\mathrm{cost}_{d,\mathbb{R}} - \mathrm{cost}_{d,\mathbb{C}} = d-2$ for each nonreal conjugate pair, independent of polynomial degree).

Formalized analysis and nonstandard analysis. Banach-space calculus, Bochner integration, and normed-algebra exponentials in Lean's Mathlib [16] provide the infrastructure for `Semigroup.lean`, while Isabelle/HOL's multivariate analysis library [8] and Fleuriot's formalization of nonstandard analysis [5] (`HOL-Nonstandard_Analysis`) underpin `Semigroup_NSA.thy`. Section 4 provides a controlled side-by-side comparison of standard and nonstandard formalizations of the same theorem across both systems.

2 Definitions, Surjections, and the Two-Dimensional Core

2.1 Multi-way separable rank and exponential-polynomial representations

Throughout the paper, we maintain a strict profile-notation convention: $\varphi : \mathbb{R} \to \mathbb{R}$ always denotes a *real-valued* radial profile, $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$ denotes its complexification, $\psi : \mathbb{R} \to \mathbb{C}$ denotes a general *complex-valued* radial profile, and $f : \mathbb{R} \to \mathbb{K}$ denotes a profile over an arbitrary field or commutative semiring $\mathbb{K}$. Let ι be a finite index type and $(X_i)_{i\in\iota}$ a family of sets. In `Defs.lean`:

Definition 2.1 (`IsSumOfProducts`, `separableRank`)**.** A kernel $K : (\prod_{i\in\iota} X_i) \to \mathbb{K}$ is a *sum of N products* if there exist univariate factors $f : \mathrm{Fin}\, N \to \prod_{i\in\iota}(X_i \to \mathbb{K})$ such that

$$K(x) = \sum_{n\in\mathrm{Fin}\, N} \prod_{i\in\iota} f_n(i, x_i) \qquad \text{for all } x \in \prod_{i\in\iota} X_i.$$

Its *separable rank* is $\mathrm{separableRank}(K) = \inf\{N \in \mathbb{N} : \mathrm{IsSumOfProducts}(K, N)\} \in \mathbb{N} \cup \{\infty\}$ (`ENat`, written `Nat`-with-top in Mathlib).

Separable rank is subadditive under pointwise sums (`IsSumOfProducts.finsetSum`) and monotone under coordinatewise composition (`IsSumOfProducts.comp`). Crucially, coordinatewise *surjections* preserve separable rank exactly (`separableRank_comp_surjective`): if $g_i : Y_i \to X_i$ has a right inverse for each $i \in \iota$, then

$$\mathrm{separableRank}\big(y \mapsto K(i \mapsto g_i(y_i))\big) = \mathrm{separableRank}(K). \tag{2}$$

Since $x \mapsto x^2$ maps $\mathbb{R}$ onto $\mathbb{R}_{\geq 0} = [0, \infty)$, applying (2) to $\iota = \mathrm{Fin}\, d$ identifies the d-way separable rank of any radial kernel with that of the d-way additive Hankel kernel (`separableRank_radial_eq_hankel` in `Translations.lean`):

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = \mathrm{separableRank}\left(s \in \mathbb{R}^d_{\geq 0} \mapsto \varphi\left(\sum_{j=0}^{d-1} s_j\right)\right). \tag{3}$$

The same lemma immediately transfers every theorem below to axis-aligned ellipsoidal kernels $\varphi(\sum_j x_j^2/a_j^2)$ ($a_j \neq 0$) and ℓ_p superellipsoidal kernels $\varphi(\sum_j |x_j|^p)$ ($p > 0$). Moreover, for $d \geq 2$, padding $(y_0, y_1) \in \mathbb{R}^2$ with $d-2$ zeros (`padTwo`) yields the slice monotonicity bound (`separableRank_radial_two_le`):

$$\mathrm{separableRank}\big(y \in \mathbb{R}^2 \mapsto \varphi(y_0^2 + y_1^2)\big) \leq \mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right). \tag{4}$$

To state real-valued classifications without redundant conjugate duplicates, a *component* $c = (P, \lambda)$ (`Component`) pairs a polynomial $P \in \mathbb{C}[X]$ with a complex rate $\lambda \in \mathbb{C}$ and evaluates on $\mathbb{R}$ as $c.\mathrm{eval}(s) = \mathrm{Re}(P(s)e^{\lambda s})$, with profile $\mathrm{profile}(\mathrm{cs}, s) = \sum_{c \in \mathrm{cs}} c.\mathrm{eval}(s)$.

Definition 2.2 (`Component.WellFormed`, `IsRepresentation`, `multiTotalCost`)**.** A component $c = (P, \lambda)$ is *well-formed* if $P \neq 0$, $\mathrm{Im}\,\lambda \geq 0$, and $\mathrm{Im}\,\lambda = 0 \implies P \in \mathbb{R}[X]$ (every coefficient of P has zero imaginary part). A list cs is a *representation* (`IsRepresentation`) if all its components are well-formed and their rates λ are pairwise distinct. In dimension $d \geq 2$, the d-way rendering cost of c (`Component.multiCost`) is

$$c.\mathrm{multiCost}(d) = \begin{cases} (d-1)\deg P + 1 & \text{if } \mathrm{Im}\,\lambda = 0, \\ 2(d-1)\deg P + d & \text{if } \mathrm{Im}\,\lambda > 0, \end{cases}$$

with list sum $\mathrm{multiTotalCost}(d, \mathrm{cs}) = \sum_{c \in \mathrm{cs}} c.\mathrm{multiCost}(d)$. At $d = 2$, $c.\mathrm{multiCost}(2) = c.\mathrm{cost}$ ($\deg P + 1$ for real rates, $2(\deg P + 1)$ for nonreal rates, summed in `totalCost`). A representation is *Gaussian* (`IsGaussianRepresentation`) if every $c \in \mathrm{cs}$ has $\deg P = 0$, where $\mathrm{multiTotalCost}(d, \mathrm{cs}) = \mathrm{gaussianTotalCost}(d, \mathrm{cs}) = r + dc$ for r real and c nonreal components.

Remark 2.3 (Bijection with conjugate-symmetric complex exponential polynomials)**.** Why do the conditions $\mathrm{Im}\,\lambda \geq 0$ and $(\mathrm{Im}\,\lambda = 0 \implies P \in \mathbb{R}[X])$ make real representations canonical? Because $\mathrm{Re}(\overline{P}(s)e^{\bar{\lambda}s}) = \mathrm{Re}(P(s)e^{\lambda s})$ (where $\overline{P}$ conjugates coefficients) and $\mathrm{Re}(iae^{\mu s}) = 0$ for $a, \mu \in \mathbb{R}$, unrestricted pairs (P, λ) over-parameterize real profiles. Under `Component.WellFormed`, each real rate ($\mathrm{Im}\,\lambda = 0, P \in \mathbb{R}[X]$) expands in $\mathbb{C}$ to a single term $P(s)e^{\lambda s}$, whereas each upper-half-plane rate ($\mathrm{Im}\,\lambda > 0$) expands via $\mathrm{Re}\,z = \frac{1}{2}(z + \bar{z})$ to the conjugate pair $\frac{1}{2}P(s)e^{\lambda s} + \frac{1}{2}\overline{P}(s)e^{\bar{\lambda}s}$ at distinct rates $\lambda \neq \bar{\lambda}$ (`injective_rates_with_conjugates`). Since exponential polynomials at distinct complex rates are $\mathbb{C}$-linearly independent on $(0, \infty)$ (`eq_zero_of_sum_expPoly_eq_zero'` in `ExpPoly.lean`), a continuous real profile determines its unordered set of well-formed components uniquely. Notice also that for $\mathrm{Im}\,\lambda > 0$, the complex rank of this conjugate pair by Theorem 3.1 is $2((d-1)\deg P + 1) = 2(d-1)\deg P + 2$, so $c.\mathrm{multiCost}(d) - 2((d-1)\deg P + 1) = d - 2$: every nonreal conjugate pair incurs an exact real-over-complex rank penalty of $d - 2$, independent of $\deg P$.

2.2 Two-coordinate row span, complexification, and basis-free necessity

When $\iota = \mathrm{Fin}\,2$ and $\mathbb{K}$ is a field, `RowSpan.lean` defines $\mathrm{rowSpan}(K) \leq (X \to \mathbb{K})$ as the $\mathbb{K}$-linear span of the sections $\{x \mapsto K(x, y) : y \in X\}$ (fixing the second coordinate and varying the first) and proves that $\mathrm{separableRank}(K) < \infty$ holds if and only if $\mathrm{rowSpan}(K)$ is finite-dimensional over $\mathbb{K}$, in which case $\mathrm{separableRank}(K) = \mathrm{finrank}_{\mathbb{K}}(\mathrm{rowSpan}\,K)$ (`separableRank_eq_finrank_rowSpan`). For the two-variable Hankel kernel $(s_0, s_1) \mapsto f(s_0 + s_1)$ on $\mathbb{R}^2_{\geq 0}$ ($f : \mathbb{R} \to \mathbb{K}$), the row span is literally the *translate span*

$$\mathcal{W}(f) = \mathrm{span}_{\mathbb{K}}\{u \in \mathbb{R}_{\geq 0} \mapsto f(u + t) : t \in \mathbb{R}_{\geq 0}\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K}).$$

Scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ (`Complexification.lean`) gives the bound $\mathrm{separableRank}_{\mathbb{C}}(K_{\mathbb{C}}) \leq \mathrm{separableRank}_{\mathbb{R}}(K)$ for every finite index set ι, where $K_{\mathbb{C}}(x) = (K(x) : \mathbb{C})$ (`separableRank_ofReal_le`). In *two* coordinates ($\iota = \mathrm{Fin}\,2$), the reverse inequality also holds (`separableRank_ofReal`): $\mathbb{R}$-linearly independent rows in $\mathbb{R}^X$ remain $\mathbb{C}$-linearly independent in $\mathbb{C}^X$ (tested by taking real and imaginary parts of any $\mathbb{C}$-linear relation), so $\mathrm{finrank}_{\mathbb{C}}\,\mathrm{rowSpan}(K_{\mathbb{C}}) = \mathrm{finrank}_{\mathbb{R}}\,\mathrm{rowSpan}(K)$, including when both ranks are ∞.

Necessity and the two-dimensional lower bound share a single coordinate-free realization in `TranslationSemigroup.lean`:

1. Suppose $\psi : \mathbb{R} \to \mathbb{C}$ is continuous on $[0, \infty)$ and $\dim_{\mathbb{C}} \mathcal{W}(\psi) < \infty$. Equipping $\mathcal{W}(\psi)$ with a norm transferred across a linear equivalence $\mathcal{W}(\psi) \simeq_{\mathbb{C}} \mathbb{C}^n$ makes $\mathcal{W}(\psi)$ a finite-dimensional

Table 1: Glossary of profile notation and closed-form d-way separable rank costs ($d \geq 2$).

Symbol / Lean identifier	Domain / definition	Exact d-way rank / specialization
φ, $\varphi_{\mathbb{C}}$, ψ	$\varphi : \mathbb{R} \to \mathbb{R}$; $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$; $\psi : \mathbb{R} \to \mathbb{C}$	Real profile, complexification, complex profile
$\mathcal{W}(f)$ (`translateSpan`)	$\mathrm{span}_{\mathbb{K}}\{u \mapsto f(u+t) : t \geq 0\} \leq (\mathbb{R}_{\geq 0} \to \mathbb{K})$	Equals separableRank$(x \in \mathbb{R}^2 \mapsto f(x_0^2 + x_1^2))$
$c = (P, \lambda)$ (`Component`)	$c.\mathrm{eval}(s) = \mathrm{Re}(P(s)e^{\lambda s})$, $k = \deg P$	$P \neq 0$, $\mathrm{Im}\,\lambda \geq 0$, $\mathrm{Im}\,\lambda{=}0 \Rightarrow P \in \mathbb{R}[X]$
Complex term $P_\mu(s)e^{\mu s}$	Nonzero $P_\mu \in \mathbb{C}[X]$ at distinct rates $\mu \in S$	$\mathrm{cost}_{d,\mathbb{C}} = (d-1)k + 1$ (Theorem 3.1)
c.multiCost(d) ($\mathrm{Im}\,\lambda = 0$)	Real-rate component ($P \in \mathbb{R}[X]$, degree k)	$(d-1)k + 1$ real products ($= \mathrm{cost}_{d,\mathbb{C}}$)
c.multiCost(d) ($\mathrm{Im}\,\lambda > 0$)	Nonreal conjugate pair ($P \in \mathbb{C}[X]$, degree k)	$2(d-1)k + d$ real products ($= \mathrm{cost}_{d,\mathbb{C}} + d - 2$)
`multiTotalCost`(d, cs)	$\sum_{c \in \mathrm{cs}} c.\mathrm{multiCost}(d)$ for a representation cs	Exact d-D real radial rank (Theorem 3.5)
`totalCost`(cs)	Specialization multiTotalCost$(2, \mathrm{cs})$ ($d = 2$)	$(k+1)$ per real rate, $2(k+1)$ per nonreal rate
`gaussianTotalCost`(d, cs)	Specialization of `multiTotalCost` to $\deg P = 0$	$r + dc$ (r real, c nonreal Gaussians)

Banach space whose induced topology agrees with the subspace topology inherited from the product topology on $\mathbb{R}_{\geq 0} \to \mathbb{C}$ (since all Hausdorff vector topologies on $\mathbb{C}^n$ coincide).

2. Because $\mathcal{W}(\psi)$ is invariant under non-negative shifts $(T_t f)(u) = f(u+t)$ ($t \geq 0$), continuity of ψ gives pointwise continuity of $t \mapsto T_t f$ on each generator and hence operator-norm continuity of the semigroup $t \mapsto T_t \in \mathcal{L}(\mathcal{W}(\psi))$ on $[0, \infty)$ (`continuous_shiftCLM`).

3. Applying the Banach-algebra semigroup theorem (`Semigroup.lean`, Section 4) yields a generator $A \in \mathcal{L}(\mathcal{W}(\psi))$ with $T_t = \exp(tA)$ for all $t \geq 0$, so $\psi(t) = \mathrm{eval}_0(\exp(tA)\psi_0)$ where $\psi_0 = \psi|_{\mathbb{R}_{\geq 0}} \in \mathcal{W}(\psi)$ (`profile_of_generator`).

4. Decomposing ψ_0 into generalized eigenspaces $\ker((A - \mu I)^k)$ of A (`Module.End.iSup_maxGenEigenspace_eq_top` in `Necessity.lean`) truncates the exponential series of $(A - \mu I)$ on each summand, expressing $\psi(t) = \sum_{\mu \in S} Q_\mu(t)e^{\mu t}$ for $t \geq 0$. When $\psi = \varphi_{\mathbb{C}}$ comes from a real profile φ, averaging $\varphi(t) = \frac{1}{2}(\varphi(t) + \overline{\varphi(t)})$ and pairing μ with $\bar{\mu}$ using exponential-polynomial uniqueness on $(0, \infty)$ (`ExpPoly.lean`, `RealExpPoly.lean`) regroups the sum into a well-formed real representation (`Representation.lean`).

For the two-dimensional exact rank and annihilator characterization (`Differential.lean`, `ComplexRank.lean`, `Annihilator.lean`), let D denote differentiation within $[0, \infty)$ (right differentiation at 0). Applying Cayley–Hamilton to the generator A on $\mathcal{W}(\psi)$ shows that the monic characteristic polynomial $\chi_A(X)$ of degree $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ annihilates every orbit $t \mapsto \ell(\exp(tA)v)$, so $\chi_A(D)\psi = 0$ on $[0, \infty)$. Conversely, for $\psi(t) = \sum_i P_i(t)e^{\lambda_i t}$ with distinct rates λ_i and $P_i \neq 0$, Hasse-derivative independence shows that $q(D)\psi = 0$ on $[0, \infty)$ if and only if the product annihilator

$$p(X) = \prod_i (X - \lambda_i)^{\deg P_i + 1}$$

divides $q(X)$ (`differentialOperator_sum_expPoly_eq_zero_iff`). Thus $p \mid \chi_A$, giving $\sum_i(\deg P_i + 1) = \deg p \leq \deg \chi_A = \dim_{\mathbb{C}} \mathcal{W}(\psi)$, while Taylor shift expansion of $P_i(u+t)e^{\lambda_i(u+t)}$ gives the matching upper bound $\dim_{\mathbb{C}} \mathcal{W}(\psi) \leq \sum_i(\deg P_i + 1)$. Table 1 summarizes the notation and cost formulas used throughout Sections 3–5.

3 Exact Multi-Way Separable Ranks in $d \geq 2$ Dimensions

In dimension $d \geq 3$, there is no single row-span dimension equal to separableRank(K), and scalar extension $\mathbb{R} \hookrightarrow \mathbb{C}$ no longer preserves rank. Instead, `MultiComplexRank.lean` and `MultiRealRank.lean` follow a four-step architecture linking upper and lower bounds:

Radial $K(x) = f(\|x\|^2)$ on $\mathbb{R}^d$ $\xleftarrow[\text{surjection } s_j = x_j^2]{\text{Eq. (3)}}$ Hankel $(s_0, \dots, s_{d-1}) \mapsto f\left(\sum_{j=0}^{d-1} s_j\right)$ on $\mathbb{R}^d_{\geq 0}$

Constructive upper bounds ($\leq N$)
- $\mathbb{C}$: $L = (d-1)k+1$ roots of unity (Sec. 3.2)
- $\mathbb{R}$, $\operatorname{Im}\lambda{=}0$: Lagrange at $0, \dots, L-1$ (Sec. 3.3)
- $\mathbb{R}$, $\operatorname{Im}\lambda{>}0$: $R = 2(d-1)k+d$ roots (Prop. 3.4)

Multilinear lower bounds ($\geq N$)
- Dual span of $v(\mathbb{R}_{\geq 0}) \Rightarrow$ linear $\ell_{n,j} \in V^*$ (Sec. 3.1)
- Peel $c_i{=}1$ by 1 ($\mathbb{C}, \mathbb{R}_{\text{real}}$) or d ($\mathbb{R}_{\text{nonreal}}$)
- Peel $c_i \geq 2$ by $d-1$ or $2(d-1)$ via $+1$ shift

3.1 Bridging function spaces and multilinear tensor rank

Suppose V is a $\mathbb{K}$-vector space over a field $\mathbb{K}$, $v : \alpha \to V$ is an evaluation map whose image spans V ($\operatorname{span}_{\mathbb{K}}(\operatorname{range} v) = \top$), and $M : V^d \to \mathbb{K}$ is a d-multilinear map (`MultilinearMap` $\mathbb{K}$ (Fin $d \to V$) $\mathbb{K}$). Lemma `exists_linear_factors_of_span_eq_top` proves that if the composed kernel $s \in \alpha^d \mapsto M(v(s_0), \dots, v(s_{d-1}))$ admits a functional decomposition (1) of rank R with arbitrary set-theoretic factors $f_{n,j} : \alpha \to \mathbb{K}$, then there exist *linear* functionals $\ell_{n,j} \in V^*$ such that

$$M(u_0, \dots, u_{d-1}) = \sum_{n=0}^{R-1} \prod_{j=0}^{d-1} \ell_{n,j}(u_j) \qquad \text{for all } u_0, \dots, u_{d-1} \in V. \tag{5}$$

Indeed, selecting a $\mathbb{K}$-basis $\{v(s_b)\}_{b \in B}$ of V from $\operatorname{range}(v)$ defines unique linear functionals $\ell_{n,j} \in V^*$ satisfying $\ell_{n,j}(v(s_b)) = f_{n,j}(s_b)$; both sides of (5) are d-multilinear maps that agree on every d-tuple of basis vectors, so they agree on V^d. To verify $\operatorname{span}_{\mathbb{K}}(\operatorname{range} v) = \top$ when $\dim_{\mathbb{K}} V < \infty$, Lemma `span_range_eq_top_of_dual` shows by quotient duality that it suffices to check $L \circ v = 0 \implies L = 0$ for every $L \in V^*$.

3.2 Exact d-way complex exponential-polynomial rank

Fix $d \geq 2$, a finite index set ι, injective rates $\lambda : \iota \to \mathbb{C}$, and nonzero polynomials $P_i \in \mathbb{C}[X]$ of degree $k_i = \deg P_i$ ($i \in \iota$).

Upper bound via shifted root-of-unity interpolation. For a single polynomial $P \in \mathbb{C}[X]$ of degree $k = \deg P$, put $L = (d-1)k + 1 \geq 1$ and $\zeta = \exp(2\pi i/L)$. Using the q-th Hasse derivative $D^{[q]}P$ (so $P(s_0 + t) = \sum_{q=0}^{k} (D^{[q]}P)(s_0)\, t^q$), define the reversed Hasse polynomial on coordinate 0 and the truncated exponential polynomials on coordinates $j \in \{1, \dots, d-1\}$ (`revHassePoly`, `truncExpPoly`):

$$R(s_0; X) = \sum_{q=0}^{k} q!\, (D^{[q]}P)(s_0)\, X^{k-q}, \qquad E_k(s_j; X) = \sum_{q=0}^{k} \frac{s_j^q}{q!} X^q,$$

$$Q(X) = R(s_0; X) \prod_{j=1}^{d-1} E_k(s_j; X).$$

Binomial convolution gives $[X^q] \prod_{j=1}^{d-1} E_k(s_j; X) = (s_1 + \cdots + s_{d-1})^q / q!$ for all $0 \leq q \leq k$, so the coefficient of X^k in $Q(X)$ is precisely $P(s_0 + s_1 + \cdots + s_{d-1})$. Crucially, because $d \geq 2$, we have $k < L$ and $\deg Q \leq k + (d-1)k = k + L - 1 < k + L$. For *any* polynomial Q with $\deg Q < k + L$ and $k < L$, discrete Fourier averaging over only L roots of unity extracts $[X^k]Q(X)$ (`coeff_eq_sum_rootsOfUnity`):

$$[X^k]Q(X) = \frac{1}{L} \sum_{r=0}^{L-1} \zeta^{-rk}\, Q(\zeta^r),$$

because every monomial index $n \in \{0, \dots, k+L-1\}$ satisfies $-L < -k \leq n-k < L$, so $L \mid (n-k)$ holds if and only if $n = k$. Multiplying the j-th factor of $Q(\zeta^r)$ by $e^{\lambda s_j}$ (and absorbing $\frac{1}{L}\zeta^{-rk}$ into coordinate 0) expresses $P(\sum_{j=0}^{d-1} s_j) e^{\lambda \sum_{j=0}^{d-1} s_j}$ as a sum of $L = (d-1)\deg P + 1$ coordinatewise products (`isSumOfProducts_expPoly`).

Lower bound via Alder–Strassen degree peeling. Let $m_0(i) = \deg P_i + 1$ and form the coefficient evaluation space $V = \prod_{i\in\iota} \mathbb{C}^{m_0(i)}$ with $v(s)_{i,q} = \frac{s^q}{q!}e^{\lambda_i s}$ $(0 \le q < m_0(i))$. By univariate exponential-polynomial uniqueness (`eq_zero_of_sum_expPoly_eq_zero'`), no nonzero functional in V^* vanishes on range(v), so $\operatorname{span}_{\mathbb{C}}(\operatorname{range} v) = \top$ (`span_expMonomial_eq_top`). Under v, the Hankel kernel corresponds to the d-multilinear truncated-convolution form (`truncForm`) with cutoffs $c_i \le m_0(i)$,

$$\mathcal{T}_c(u^{(0)},\dots,u^{(d-1)}) = \sum_{i\in\iota} \sum_{q_0+\cdots+q_{d-1}<c_i} w_i(q_0+\cdots+q_{d-1}) \prod_{j=0}^{d-1} u^{(j)}_{i,q_j},$$

where $w_i(c_i-1) \neq 0$ whenever $c_i > 0$. Lemma `truncForm_rank_ge` proves

$$\sum_{i:\, c_i>0} \big((d-1)(c_i-1)+1\big) \le R$$

for every rank-R decomposition (5) by strong induction on $S = \sum_{i\in\iota} c_i$:

- *Case $c_{i_0} = 1$:* Evaluating $\mathcal{T}_c$ at $(e_{i_0,0},\dots,e_{i_0,0})$ gives $w_{i_0}(0) \neq 0$, selecting an index n_0 with $\ell_{n_0,0}(e_{i_0,0}) \neq 0$. Let $\pi_{\neq i_0} : V \to V$ zero out block i_0. Substituting $\varphi_0(x) = \pi_{\neq i_0}(x) - \frac{\ell_{n_0,0}(\pi_{\neq i_0}(x))}{\ell_{n_0,0}(e_{i_0,0})} e_{i_0,0}$ on mode 0 and $\pi_{\neq i_0}$ on modes $j \ge 1$ annihilates $\ell_{n_0,0}\circ\varphi_0$ while reducing c_{i_0} from 1 to 0 (since $d \ge 2$ zeroes block i_0 on mode 1), leaving at most $R-1$ terms.
- *Case $c_{i_0} \ge 2$:* Set $s_{\mathrm{top}} = e_{i_0,c_{i_0}-1}$ and $s_0 = e_{i_0,0}$ $(s_{\mathrm{top}} \neq s_0)$. Because $(c_{i_0}-1)+(c_{i_0}-1) \ge c_{i_0}$, any rank-1 perturbation of the form $\psi(x) = x - \gamma(x)s_{\mathrm{top}}$ leaves $\mathcal{T}_c$ unchanged whenever another coordinate equals s_{top}. Stepping through modes $j = 0,\dots,d-2$ sequentially evaluates $\mathcal{T}_c$ with $u^{(j)} = s_{\mathrm{top}}$ and $u^{(k)} = \psi_k(s_0)$ $(k \neq j)$ to find $d-1$ distinct indices $D \subseteq \operatorname{Fin} R$ $(|D| = d-1)$ killed on modes $0,\dots,d-2$ by rank-1 perturbations ψ_j along s_{top}. On modes $0,\dots,d-2$ we then truncate block i_0 to indices $q < c_{i_0}-1$ before applying ψ_j, while on mode $d-1$ we shift block i_0's degree indices up by $+1$ $(e_{i_0,q} \mapsto e_{i_0,q+1}$ for $q < c_{i_0}-1)$: this transforms $\mathcal{T}_c$ into the reduced form $\mathcal{T}_{c'}$ $(c' = c[i_0 \mapsto c_{i_0}-1])$ with shifted weights $w'_{i_0}(m) = w_{i_0}(m+1)$—preserving the nonzero top weight $w'_{i_0}(c'_{i_0}-1) = w_{i_0}(c_{i_0}-1) \neq 0$—while dropping all $d-1$ terms in D.

Combining this lower bound at $c = m_0$ with the upper bound and Section 2 gives:

Theorem 3.1 (d-D Complex Radial Rank, `separableRank_radial_complex_eq_iff`)**.** *Let $d \ge 2$ and let $\psi : \mathbb{R} \to \mathbb{C}$ be continuous on $[0,\infty)$. For every $N \in \mathbb{N}$,*

$$\operatorname{separableRank}\left(x \in \mathbb{R}^d \mapsto \psi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = N$$

if and only if there exist a finite set $S \subset \mathbb{C}$ and polynomials $(P_\mu)_{\mu\in S}$ with $P_\mu \neq 0$ such that $\psi(s) = \sum_{\mu\in S} P_\mu(s)e^{\mu s}$ for all $s \ge 0$ and $\sum_{\mu\in S}\big((d-1)\deg P_\mu + 1\big) = N$.

3.3 Exact d-way real rank for arbitrary degrees (`MultiRealRank.lean`)

Before treating polynomial degrees, note that for constant factors $(d \ge 1)$, expanding $\operatorname{Re}(\prod_{j=0}^{d-1} z_j)$ into real and imaginary parts gives 2^{d-1} real products, whereas d real products always suffice:

Proposition 3.2 (Alternating $2d$-th root-of-unity identity, `re_prod_eq_sum_re`)**.** *For every integer $d \ge 1$ and all $z_0,\dots,z_{d-1} \in \mathbb{C}$,*

$$\operatorname{Re}\left(\prod_{j=0}^{d-1} z_j\right) = \frac{2^{d-1}}{d} \sum_{m=0}^{d-1} (-1)^m \prod_{j=0}^{d-1} \operatorname{Re}\Big(z_j e^{-i\pi m/d}\Big). \tag{6}$$

Proof. Write $\mathrm{Re}(z_j e^{-i\pi m/d}) = \frac{1}{2}\left(\overline{z_j} e^{i\pi m/d} + z_j e^{-i\pi m/d}\right)$ and expand over subsets $S \subseteq \{0,\dots,d-1\}$: multiplying by $(-1)^m = e^{i\pi m}$ yields the geometric progression w_S^m with $w_S = e^{2\pi i|S|/d}$ ($w_S^d = 1$), whose sum over $0 \le m < d$ equals d for $S \in \{\emptyset, \{0,\dots,d-1\}\}$ and 0 for $0 < |S| < d$. □

Absorbing $\frac{2^{d-1}}{d}(-1)^m$ into coordinate 0 shows that the real part of a rank-N complex d-way kernel has real rank at most Nd (`IsSumOfProducts.re`), establishing qualitative d-D classification:

Theorem 3.3 (d-D Finite-Rank Classification, `separableRank_radial_lt_top_iff_exists`)**.** *Let $d \ge 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0,\infty)$. Then*

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) < \infty$$
$$\iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge \forall s \ge 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s).$$

For degree $k = \deg P \ge 1$, however, multiplying Theorem 3.1's complex rank $(d-1)k+1$ by d via `IsSumOfProducts.re` gives $d(d-1)k+d$, which is non-optimal. Instead, sharp upper bounds differ for real and nonreal rates (`isSumOfProducts_radial_multiComponent`):

- *Real rates ($\mathrm{Im}\,\lambda = 0$, $P \in \mathbb{R}[X]$ of degree k):* On coordinates $1,\dots,d-1$, the real polynomial $A(X) = \prod_{j=1}^{d-1} E_k(x_j^2; X) \in \mathbb{R}[X]$ has $\deg A \le (d-1)k < L := (d-1)k+1$. Real Lagrange interpolation (`Lagrange.eq_interpolate`) at the L real nodes $0, 1, \dots, L-1 \in \mathbb{R}$ expresses each $[X^q]A(X) = (\sum_{j=1}^{d-1} x_j^2)^q/q!$ $(0 \le q \le k)$ as an $\mathbb{R}$-linear combination of $\prod_{j=1}^{d-1} E_k(x_j^2; r)$, yielding $(d-1)k+1$ real d-way products.
- *Nonreal rates ($\mathrm{Im}\,\lambda > 0$, $P \in \mathbb{C}[X]$ of degree k):* We generalize Proposition 3.2 from scalars to polynomials of degree at most k ($d \ge 2$):

Proposition 3.4 (Universal root-of-unity polynomial-product identity, `re_coeff_prod_eq_sum_re`)**.** *Let $d \ge 2$, $k \ge 0$, and $R = 2(d-1)k + d$. For any $U_0, \dots, U_{d-1} \in \mathbb{C}[X]$ with $\deg U_j \le k$,*

$$\mathrm{Re}\left([X^k]\prod_{j=0}^{d-1} U_j(X)\right) = \frac{2^{d-1}}{R}\sum_{r=0}^{R-1}(-1)^r \prod_{j=0}^{d-1} \mathrm{Re}\left(U_j\left(e^{\frac{2\pi i r}{R}}\right) e^{-\frac{i\pi r(2k+1)}{R}}\right). \tag{7}$$

Proof. Expanding $\mathrm{Re}\, w = \frac{1}{2}(\bar{w}+w)$ over subsets $S \subseteq \{0,\dots,d-1\}$ of size $s = |S|$ and coefficient indices $q \in \{0,\dots,k\}^d$, and noting that $(-1)^r = e^{i\pi r R/R}$ with $R = (2k+1)d - 2k$, gives the r-th phase $\exp(2\pi i r K(S,q)/R)$ where

$$K(S,q) = (2k+1)s + \sum_{j \notin S} q_j - \sum_{j \in S} q_j - k \in \mathbb{Z}.$$

Because $0 \le q_j \le k$ and $d \ge 2$: (i) when $S = \emptyset$ ($s = 0$), $-R < -k \le K(\emptyset, q) = \sum_j q_j - k \le (d-1)k < R$, so $R \mid K(\emptyset, q) \iff \sum_j q_j = k$; (ii) when $S = \{0,\dots,d-1\}$ ($s = d$), $K = R + (k - \sum_j q_j)$, so $R \mid K \iff \sum_j q_j = k$; and (iii) when $1 \le s \le d-1$, $K(S,q) \ge (2k+1)s - sk - k = (k+1)s - k \ge 1$ and $K(S,q) \le (2k+1)s + (d-s)k - k = (k+1)s + (d-1)k \le 2(d-1)k + d - 1 = R - 1$, so $R \nmid K(S,q)$ and the geometric sum over $0 \le r < R$ vanishes identically. Applying (7) to $U_0(X) = e^{\lambda x_0^2} R(x_0^2; X)$ and $U_j(X) = e^{\lambda x_j^2} E_k(x_j^2; X)$ $(j \ge 1)$ yields $2(d-1)k + d$ real d-way products. □

Unified Alder–Strassen lower bound over $\mathbb{R}$ (`realExpForm_rank_ge`). Let $G = (\prod_{i\in\iota_{\mathbb{R}}} \mathbb{R}^{m_{\mathbb{R}}(i)}) \times (\prod_{k\in\iota_{\mathbb{C}}} \mathbb{C}^{m_{\mathbb{C}}(k)})$ with $m(i) = \deg P_i + 1$. Conjugate-rate uniqueness shows via duality that the exponential-monomial curve $s \mapsto ((s^q/q!\, e^{\mu_i s}), (s^q/q!\, e^{\lambda_k s}))$ spans G over $\mathbb{R}$ (`span_realExpEval_eq_top`), transferring any rank-R decomposition of the Hankel kernel to $\mathbb{R}$-linear functionals $\ell_{n,j} \in G^*$ for the combined truncated-convolution form `realExpForm`. Note first that every $L \in \mathrm{Hom}_{\mathbb{R}}(\mathbb{C}, \mathbb{R})$ has a nonzero kernel vector $w \in \ker L \setminus \{0\}$ (`exists_ker_ne_zero`), from which induction on $d \geq 2$ gives $d \leq R$ for any single nonzero complex product $\mathrm{Re}(\beta \prod_{j=0}^{d-1} z_j)$ (`single_complex_rank_ge`). Strong induction on the cutoff sum then peels:

1. *Real cutoffs (`realExpForm_rank_ge`):* Identical to Section 3.2 (1 term when $c_{\mathbb{R}}(i_0) = 1$, $d-1$ terms when $c_{\mathbb{R}}(i_0) \geq 2$), while keeping the $\mathbb{C}$-coordinates of G fixed so all nonreal blocks are untouched.
2. *Nonreal cutoff $c_{\mathbb{C}}(k_0) = 1$ (`cplxTruncForm_rank_ge`):* On mode 0, evaluating at 1_{k_0} and at $w_1 \in \ker(\ell_{n_1,0}|_{\mathbb{C}e_{k_0,0}}) \setminus \{0\}$ kills 2 distinct terms, and on each mode $j = 1, \ldots, d-2$, `single_complex_rank_ge` forces one additional active term outside the killed set D (else $|D| = j+1 < d$ terms would represent $\mathrm{Re}(\beta_{k_0} \prod z_q)$), reaching $|D| = d$ before zeroing block k_0 on mode $d-1$.
3. *Nonreal cutoff $c_{\mathbb{C}}(k_0) \geq 2$:* Let $q_{\mathrm{top}} = c_{\mathbb{C}}(k_0) - 1 \geq 1$ with weight $\beta_{\mathrm{top}} \neq 0$. In each mode $j = 0, \ldots, d-2$, evaluating at $u(z)$ with $\mathrm{emb}_{\mathrm{top}}(z)$ on mode j, $\mathrm{emb}_0(\overline{\beta_{\mathrm{top}} z})$ on mode $d-1$, and $\psi_m(\mathrm{emb}_0(1))$ elsewhere gives $|\beta_{\mathrm{top}} z|^2 > 0$ for every $z \in \mathbb{C} \setminus \{0\}$ (since $2q_{\mathrm{top}} \geq c_{\mathbb{C}}(k_0)$ kills all top-degree cross-terms). Testing $z = 1$ and $z = w_1 \in \ker(\ell_{n_1,j} \circ \mathrm{emb}_{\mathrm{top}}) \setminus \{0\}$ kills 2 distinct terms per mode—$2(d-1)$ terms across modes $0, \ldots, d-2$—after which low-degree truncation on modes $0, \ldots, d-2$ and $+1$ index shifting on mode $d-1$ reduce $c_{\mathbb{C}}(k_0) \mapsto c_{\mathbb{C}}(k_0) - 1$ with shifted top weight $\beta_{\mathrm{top}} \neq 0$.

Theorem 3.5 (Exact d-D Real Radial Rank, `separableRank_radial_eq_iff`)**.** *Let $d \geq 2$ and let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$. For every $N \in \mathbb{N}$,*

$$\mathrm{separableRank}\left(x \in \mathbb{R}^d \mapsto \varphi\left(\sum_{j=0}^{d-1} x_j^2\right)\right) = N$$

$$\iff \exists\, \mathrm{cs},\ \mathrm{IsRepresentation}(\mathrm{cs}) \wedge (\forall s \geq 0,\ \varphi(s) = \mathrm{profile}(\mathrm{cs}, s)) \wedge \mathrm{multiTotalCost}(d, \mathrm{cs}) = N.$$

In particular, for a Gaussian representation with r real and c nonreal rates, $\mathrm{separableRank} = r + dc$ *(`separableRank_radial_gaussian_profile`), and in $d = 2$,* $\mathrm{separableRank} = \mathrm{totalCost}(\mathrm{cs})$ *(`separableRank_radial_two_eq_iff`).*

Theorem 3.6 (Minimal Differential Annihilator)**.** *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be continuous on $[0, \infty)$ with finite-dimensional translate span and two-dimensional radial rank* $\mathrm{separableRank}(x \in \mathbb{R}^2 \mapsto \varphi(x_0^2 + x_1^2)) = N < \infty$ *(`exists_minimal_differential_annihilator`). There exists a monic polynomial $p \in \mathbb{C}[X]$ of degree N such that for every $q \in \mathbb{C}[X]$, the half-line differential equation $q(D)\varphi_{\mathbb{C}} = 0$ holds on $[0, \infty)$ (where $\varphi_{\mathbb{C}}(s) = (\varphi(s) : \mathbb{C})$) if and only if $p \mid q$.*

Theorem 3.7 (Exact vs. Border Rank Separation)**.** *Let $d \geq 2$ and define $K_d(x) = \left(\sum_{j=0}^{d-1} x_j^2\right) \cos\left(\sum_{j=0}^{d-1} x_j^2\right)$ on $\mathbb{R}^d$. In `Main.lean`:*

1. Exact real rank: $\mathrm{separableRank}_{\mathbb{R}}(K_d) = 3d-2$, *giving* 7 *in* $d = 3$ *(`separableRank_radial_sq_mul_cos`).*
2. Exact complex rank: $\mathrm{separableRank}_{\mathbb{C}}((K_d)_{\mathbb{C}}) = 2d$, *giving* 6 *in* $d = 3$ *(`separableRank_radial_sq_mul_cos_complex`).*

3. Compact-open border rank $\leq 2d$: *Defining for $h \in \mathbb{R}$ the difference quotient*

$$F_h(x) = \frac{1}{h}\,\mathrm{Re}\left(\prod_{j=0}^{d-1}(1+hx_j^2)e^{ix_j^2}\right) - \frac{1}{h}\,\mathrm{Re}\left(\prod_{j=0}^{d-1}e^{ix_j^2}\right),$$

every F_h satisfies $\mathrm{separableRank}_{\mathbb{R}}(F_h) \leq 2d$*, and $F_h \to K_d$ uniformly on every compact subset $C \subset \mathbb{R}^d$ as $h \to 0$, $h \neq 0$ (`exists_rank_le_two_mul_tendstoUniformlyOn_radial_sq_mul_cos`). Consequently, for every $d \geq 3$, $\{F \in C(\mathbb{R}^d, \mathbb{R}) : \mathrm{separableRank}_{\mathbb{R}}(F) \leq 2d\}$ is not closed in the compact-open topology, and the real exact-minus-border rank gap is at least $d-2$.*

Proof sketch. Parts (1) and (2) instantiate Theorems 3.5 and 3.1 on $c = (X, i)$ ($\deg X = 1, \mathrm{Im}\, i = 1 > 0$) and $S = \{i, -i\}$ ($P_{\pm i} = \frac{1}{2}X$). For (3), Proposition 3.2 realizes each $\mathrm{Re}(\prod_{j=0}^{d-1}\cdots)$ in d real products, so $\mathrm{separableRank}_{\mathbb{R}}(F_h) \leq 2d$. Induction on finite products (`abs_prod_one_add_sub_le`) proves $\left|\prod_{j=0}^{d-1}(1+hx_j^2) - 1 - h\sum_{j=0}^{d-1}x_j^2\right| \leq h^2\prod_{j=0}^{d-1}(1+x_j^2)$ whenever $|h| \leq 1$, hence $|F_h(x) - K_d(x)| \leq |h| \sup_{y\in C}\prod_j(1+y_j^2) \to 0$ uniformly on C. Why can no *exact* decomposition achieve rank $2d$ at $h = 0$ when $d = 3$ (6 vs. 7)? At $h \neq 0$, F_h relies on $O(h^{-1})$ cancellation plus $O(h)$ degree-2 and degree-3 cross-terms $h(s_0s_1 + s_1s_2 + s_0s_2) + h^2 s_0s_1s_2$. Any exact decomposition at $h = 0$, by contrast, pushes forward linearly onto the truncated algebra $W = \mathbb{C}[T]/(T^2) \cong \mathbb{C} \oplus \mathbb{C}\varepsilon$ ($\varepsilon^2 = 0$), where $\mathcal{M}(u, v, w) = \mathrm{Re}(u_1v_0w_0 + u_0v_1w_0 + u_0v_0w_1)$ has zero ε^2 cross-terms: peeling $\dim_{\mathbb{R}}(0 \oplus \mathbb{C}\varepsilon)^* = 2$ terms on mode 0 and 2 terms on mode 1 leaves $\mathrm{Re}(u_0v_0w_1)$, which requires 3 more terms by `single_complex_rank_ge`, forcing $R \geq 2+2+3 = 7$. □

4 Standard vs. Nonstandard Proofs of the Semigroup Step

Every proof of necessity above rests on the continuous one-parameter semigroup theorem: if $\mathcal{A}$ is a Banach algebra over $\mathbb{R}$ and $M : \mathbb{R} \to \mathcal{A}$ satisfies $M(0) = 1$, $M(s+t) = M(s)M(t)$ for all $s, t \geq 0$, and M is continuous on $[0, \infty)$, then there exists $A \in \mathcal{A}$ such that $M(t) = \exp(tA)$ for all $t \geq 0$.

4.1 Integral smoothing vs. Euler's infinitesimal step with homogeneous scale

Integral smoothing (`Semigroup.lean`). Since $M(0) = 1$ and M is continuous at 0, there exists $\delta > 0$ such that $I_\delta = \int_0^\delta M(u)\,du$ satisfies $\|I_\delta - \delta \cdot 1\| < \delta$, making I_δ invertible in $\mathcal{A}$ by Neumann series. For $t \geq 0$, changing variables in $M(t)I_\delta = \int_t^{t+\delta} M(u)\,du$ shows by the fundamental theorem of calculus that M is right-differentiable on $[0, \infty)$ with $M'(t) = M(t)A$, where $A = (M(\delta) - 1)I_\delta^{-1}$ commutes with $M(t)$. Differentiating $t \mapsto M(t)\exp(-tA)$ gives 0, so $M(t) = \exp(tA)$ in *any* Banach algebra.

Finite differences (`GeneratorBound.lean`, `FiniteDifferenceSemigroup.lean`). Without integration, if $\|M(u) - 1\| \leq \frac{1}{2}$ on $[0, \delta]$, geometric-series telescoping bounds the difference quotients $\|(M(\delta/n) - 1)/(\delta/n)\|$. In finite dimensions, compactness extracts a convergent subsequence $(M(h_k) - 1)/h_k \to A$; along h_k, the one-sided difference quotients of $M(t)\exp(-tA)$ tend to 0, and a one-sided mean-value lemma shows $M(t)\exp(-tA) = 1$.

Nonstandard analysis with homogeneous common scale. Euler's heuristic sets $A = \mathrm{st}((M(\varepsilon) - 1)/\varepsilon)$ for a positive infinitesimal $\varepsilon > 0$ and iterates $M(\varepsilon)^{\lfloor t/\varepsilon \rfloor} \approx (1 + \varepsilon A)^{\lfloor t/\varepsilon \rfloor} \approx \exp(tA)$. Directly applying st to $(M(\varepsilon) - 1)/\varepsilon$, however, requires proving beforehand that this quotient is finite (the 124-line geometric-sum bound above).

All three nonstandard formalizations in this project eliminate that *a priori* bound by passing to *projective / homogeneous coordinates*: put $x = M(\varepsilon) - 1$ (which is infinitesimal by continuity at 0) and define the common infinitesimal scale

$$\eta = \varepsilon + \|x\| > 0. \tag{8}$$

Because $\frac{\varepsilon}{\eta} + \left\|\frac{x}{\eta}\right\| = 1$ with $\frac{\varepsilon}{\eta} \in [0,1]$ and $\left\|\frac{x}{\eta}\right\| \le 1$, both rescaled quantities are bounded by 1 *without any analytical estimate*. In a finite-dimensional algebra $\mathcal{A}$, Heine–Borel compactness gives a standard vector $B = \mathrm{st}(x/\eta)$ with $\|B\| \le 1$ and $\varepsilon/\eta \approx a := 1 - \|B\| \in [0,1]$. Now compare $N = \lfloor s/\eta \rfloor$ steps of $M(\varepsilon)$ (arriving at time $N\varepsilon \approx sa$) against N steps of $\exp(\eta B)$ (arriving at time $N\eta \approx s$) for an arbitrary standard $s \ge 0$:

$$\frac{\|M(\varepsilon) - \exp(\eta B)\|}{\eta} \le \left\|\frac{x}{\eta} - B\right\| + \left\|B - \frac{\exp(\eta B) - 1}{\eta}\right\| \approx 0.$$

Summing N single-step errors of size $\eta \cdot o(1)$ with $N\eta \le s$ yields $M(sa) = \exp(sB)$ for all standard $s \ge 0$. Could the time coordinate vanish ($a = 0$)? If $a = 0$, then $\exp(sB) = M(0) = 1$ for every $s \ge 0$; differentiating at $s = 0$ gives $B = 0$, contradicting $a + \|B\| = 1$! Hence $a > 0$, and setting $A = B/a$ with $s = t/a$ gives $M(t) = \exp(tA)$.

4.2 Three nonstandard formalizations and quantitative comparison

We formalized the homogeneous-scale nonstandard proof in three settings:

1. **Isabelle/HOL:** `Isabelle/Semigroup_NSA.thy` uses `HOL-Nonstandard_Analysis`, which provides hyperreals, hypernaturals, star extensions `*f*`, and the `transfer` proof method. Single-step error accumulation reduces to a purely algebraic noncommutative power estimate (`power_difference_bound`): if $\|u\|, \|v\| \le e^h$ with $h \ge 0$, then $\|u^n - v^n\| \le ne^{nh}\|u - v\|$, proved by induction from $u^{n+1} - v^{n+1} = u(u^n - v^n) + (u - v)v^n$. Since $\|M(\varepsilon)\| \le 1 + \|x\| \le 1 + \eta \le e^\eta$ and $\|\exp(\eta B)\| \le e^{\eta\|B\|} \le e^\eta$, `transfer` applies this ordinary inequality directly to a hypernatural step count N. (This uses $\|1\| = 1$ from Isabelle's `real_normed_algebra_1`; in Lean, $\mathcal{L}(\mathcal{W}(\psi))$ has $\|1\| = 0$ when $\psi = 0$, so the Lean theorems avoid assuming $\|1\| = 1$ and telescope steps using compact-interval bounds via `norm_replace_steps_le` instead.)
2. **Lean 4 with ultrafilter sequences:** `Experiments/NonstandardSemigroup.lean` expresses "infinitely close" directly as convergence along hyperfilter $\mathbb{N}$ (since Mathlib defines Hyperreal = Germ(hyperfilter $\mathbb{N}$) $\mathbb{R}$ without a transfer tactic), obtaining B in one line via `IsCompact.ultrafilter_le_nhds`.
3. **Lean 4 in nonstandard vocabulary:** In `Experiments/`, `Ultrapower.lean` and `UltrapowerSemigroup.lean` build a reusable 195-line ultrapower interface (`Hyp`, `std`, `ext`, $\approx_{\mathrm{h}}$, one-variable transfer lemmas) and write Euler's proof in classical nonstandard notation.

Table 2 summarizes the exact line counts. These numbers reflect *library amortization* rather than intrinsic proof complexity: `Semigroup.lean` (96 lines) is shortest and holds in every Banach algebra because Mathlib amortizes thousands of lines of Bochner integration, Neumann series, and the fundamental theorem of calculus behind compact API lemmas, whereas compactness-based proofs are inherently finite-dimensional. Across different proof styles within Lean, the 2× gap between integral smoothing (96 lines) and ultrafilter sequences (206 lines) remained stable even after eliminating the 124-line difference-quotient bound via homogeneous coordinates (8). Moreover, discounting the 195-line interface `Ultrapower.lean` as reusable library code leaves $169 + 35 = 204$ lines: without automated transfer, nonstandard vocabulary in

Table 2: Verified source sizes for the five complete proofs of the continuous one-parameter semigroup step $M(t) = \exp(tA)$. Every project helper lemma in each proof's module dependencies is counted in full (excluding standard Mathlib and Isabelle/HOL library imports).

Proof method	Prover / files	Phys.	Nonbl.	Scope
Integral smoothing (main)	Lean 4 (`Semigroup.lean`)	96	87	Any Banach algebra
Finite differences	Lean 4 (`GeneratorBound`, `FD`)	256	233	Finite-dim.
NSA (ultrafilter seq.)	Lean 4 (`NonstandardSemigroup`)	206	190	Finite-dim.
NSA (ultrapower API)	Lean 4 (`Ultrapower*` + shared)	399	344	Finite-dim.
NSA (star + `transfer`)	Isabelle (`Semigroup_NSA.thy`)	275	253	Finite-dim. ($\|1\| = 1$)

Lean merely wraps filter limits. In Isabelle/HOL, `transfer` genuinely compresses hypernatural induction, but 59 lines of local library gaps (Heine–Borel vector standard parts and half-line continuity via $M(\max(t, 0))$) plus stating transferred formulas at both standard and nonstandard types keep the total at 275 lines.

5 Applications to 2-D Blur, Steerable Filtering, and 3-D Acoustic Propagation

All numerical optimizations, finite-precision simulations, and GPU benchmarks in Sections 5.2–5.3 are implemented in reproducible scripts under `Applications/` and executed on an NVIDIA GeForce RTX 4070 Ti Super (16 GiB GDDR6X, 672 GB/s peak memory bandwidth). Before turning to specific designs, note an analytical duality linking spatial and frequency-domain radial kernels: for $\operatorname{Re}\lambda < 0$, differentiating $\mathcal{F}_{\mathbb{R}^d}[e^{\lambda\|x\|^2}](\xi) = (\pi/(-\lambda))^{d/2}\exp(\|\xi\|^2/(4\lambda))$ k times with respect to λ maps $P(\|x\|^2)e^{\lambda\|x\|^2}$ ($\deg P = k$) to $\widetilde{P}(\|\xi\|^2)e^{\|\xi\|^2/(4\lambda)}$ with $\deg\widetilde{P} = k$ and $\operatorname{Re}(1/(4\lambda)) < 0$. By Theorems 3.1 and 3.5, the d-dimensional Fourier transform preserves exact complex and real d-way separable rank across all integrable radial exponential-polynomial kernels.

5.1 Harmonic "beach-ball" differential operators and steerable kernels

Specializing Proposition 3.2 to equal symbolic factors $z_0 = \cdots = z_{d-1} = \partial_x + i\partial_y$ (writing m for the harmonic order and $D_\alpha = \cos\alpha\,\partial_x + \sin\alpha\,\partial_y$ for the directional derivative along angle α) gives, for *every* integer $m \geq 1$ (both odd and even),

$$\operatorname{Re}\big((\partial_x + i\partial_y)^m\big) = \frac{2^{m-1}}{m}\sum_{k=0}^{m-1}(-1)^k D^m_{\pi k/m}. \tag{9}$$

On a regular $2m$-point ring $\theta_n = \pi n/m$ ($n = 0, \ldots, 2m-1$) of radius $h \to 0$, antipodal points satisfy $D_{\theta_{k+m}} = -D_{\theta_k}$ while Fourier orthogonality of $(-1)^k = e^{i\pi k}$ annihilates all Taylor orders below m. For $f \in C^m$ near (x, y), (9) recovers—by elementary geometric series rather than Cauchy residues and Chebyshev partial fractions—both the odd-order ($m = 2N+1$) hexagonal/decagonal "beach-ball" operators of Niemitalo [19] and the even-order operators of Perona [22] in a single formula:

$$\lim_{h\to 0}\frac{1}{h^m}\sum_{n=0}^{2m-1}(-1)^n f\Big(x + h\cos\frac{\pi n}{m},\, y + h\sin\frac{\pi n}{m}\Big) = \frac{1}{(m-1)!\,2^{m-2}}\operatorname{Re}\big((\partial_x + i\partial_y)^m\big)f(x, y),$$

with a half-sector ring rotation $+\frac{\pi}{2m}$ yielding $+\operatorname{Im}((\partial_x + i\partial_y)^m)$. Recognizing $\partial_{\bar z} = \frac{1}{2}(\partial_x + i\partial_y)$ as the Cauchy–Riemann operator also explains why applying $\partial^m_{\bar z}$ to *any* smooth radial kernel

$\varphi(x^2+y^2) = \varphi(z\bar{z})$ produces no lower-order radial derivatives: because $\partial_{\bar{z}}(z) = 0$, induction gives

$$\left(\frac{\partial_x + i\partial_y}{2}\right)^m \varphi(x^2+y^2) = (x+iy)^m\, \varphi^{(m)}(x^2+y^2) = r^m e^{im\theta}\, \varphi^{(m)}(r^2). \tag{10}$$

When $\varphi(s)$ is a well-formed multi-Gaussian profile with r real and c nonreal upper-half-plane rates (all $\lambda_i \neq 0$, approximating, e.g., an ideal circular low-pass filter [20]), $\varphi^{(m)}(s)$ is *again* a well-formed multi-Gaussian profile at the same rates (weighted by λ_i^m) of continuous 2-way rank $R = r+2c$. Analytically differentiating the continuous-space kernel on $\mathbb{R}^2$—avoiding discrete finite-difference distortion up to Nyquist [18]—absorbs the monomials of $\mathrm{Re}((x+iy)^m)$ and $\mathrm{Im}((x+iy)^m)$ into the 1-D Gaussian factors, giving exact *continuous-domain* 2-way real separable ranks on $\mathbb{R}^2$:

$$\big(\lfloor m/2 \rfloor + 1\big)(r+2c) \quad \text{for } \cos(m\theta), \qquad \lceil m/2 \rceil (r+2c) \quad \text{for } \sin(m\theta).$$

Sampling those 1-D factors on a finite grid yields a discrete separable realization with at most that many terms (grid aliasing at special rates such as $\mathrm{Im}\,\lambda_i \in \pi\mathbb{Z}$ can collapse the sampled discrete rank further). On the 161×161 ($\omega_c = \pi/64$) orientation-estimation filter of [18], replacing direct $2 \times 161^2 = 51{,}842$ taps/pixel by a two-complex-Gaussian ($R = 4$) analytical gradient pair reduces scalar multiply–adds to 2,576 (20×, or 40× with even/odd tap-parity folding).

Continuous-domain pass-count bounds for simultaneous steerable pairs. Can sharing intermediate 1-D continuous convolutions reduce the simultaneous evaluation of $(C_m * I, S_m * I)$ below separate evaluation $2(\lfloor m/2 \rfloor + 1)R + 2\lceil m/2 \rceil R = 2(m+1)R$? No: if any two-stage 1-D schedule on $\mathbb{R}^2$ splits $(C_m, S_m) = (A_c + B_c, A_s + B_s)$ into P_y shared vertical passes followed by $\mathrm{rank}(A_c) + \mathrm{rank}(A_s)$ horizontal passes plus P_x shared horizontal passes followed by $\mathrm{rank}(B_c) + \mathrm{rank}(B_s)$ vertical passes, then

$$\begin{aligned} \dim \mathrm{rowSpan}_x(C_m, S_m) &\leq \mathrm{rank}(A_c) + \mathrm{rank}(A_s) + P_x, \\ \dim \mathrm{colSpan}_y(C_m, S_m) &\leq P_y + \mathrm{rank}(B_c) + \mathrm{rank}(B_s). \end{aligned}$$

Because C_m and S_m on $\mathbb{R}^2$ span all powers $\{x^0, \ldots, x^m\} g_n(x)$ in x and $\{y^0, \ldots, y^m\} g_n(y)$ in y, adding the two inequalities shows that *every* two-stage continuous 1-D schedule requires at least $2(m+1)R$ passes ($4R$ passes for (h_x, h_y)). Savings instead come from: (i) definite even/odd parity of every 1-D factor $u^k g_n(u)$, halving multiplications per sampled pass via $I[y+\ell] \pm I[y-\ell]$ folding; (ii) sharing $\{I *_y (y^k g_n)\}_{k=0}^{M}$ across *multiple* harmonic orders ($5R$ instead of $6R$ for $m = 0, 1$; $10R$ instead of $12R$ for $m = 1, 3$); and (iii) for the d-D gradient $(h_{x_0}, \ldots, h_{x_{d-1}})$ of a multi-Gaussian radial kernel ($d \geq 3$), recursively splitting $\{0, \ldots, d-1\}$ into halves to reduce $d^2 R_d$ passes to $T(d) R_d$ with $T(1) = 1$ and $T(d) = T(\lfloor d/2 \rfloor) + T(\lceil d/2 \rceil) + d \leq d(\lceil \log_2 d \rceil + 1)$ ($8R_3$ vs. $9R_3$ in $d = 3$; $12R_4$ vs. $16R_4$ in $d = 4$).

5.2 Worked application I: 2-D circular lens blur and $\mathrm{GL}_N(\mathbb{R})$ conditioning

In depth-of-field bokeh rendering [17, 21], the ideal defocus point-spread function is the discontinuous unit disc $\mathbf{1}_{r \leq 1}$ ($r = \sqrt{x^2+y^2}$), which has infinite separable rank: for any $0 < r_1 < \cdots < r_M < 1$, evaluating $\mathbf{1}_{x_i^2 + y_j^2 \leq 1}$ at $x_i = r_i$, $y_j = \sqrt{1 - r_j^2}$ produces an invertible $M \times M$ upper-triangular binary matrix. Introducing a transition band $[1, S]$ with $S = 1 + \tau$ ($\tau = 0.2$) and truncating each 1-D pass to $|x|, |y| \leq S$ restricts the 2-D separable support $[-S, S]^2$ to radii $r \leq \sqrt{2}\, S \approx 1.697$. For any fixed decay and oscillation rates (b_u, c_u), minimizing the sampled Chebyshev ripple $\max_{r \in [0,1] \cup [S, \sqrt{2}S]} |\varphi(r^2) - \mathbf{1}_{r \leq 1}|$ (2,000 radial nodes) over all polynomial weights is a linear program, solved with HiGHS inside heuristic differential-evolution/Nelder–Mead rate search in `Applications/blur_2d.py`.

Figure 2 and Table 3 highlight three structural consequences of Theorem 3.5:

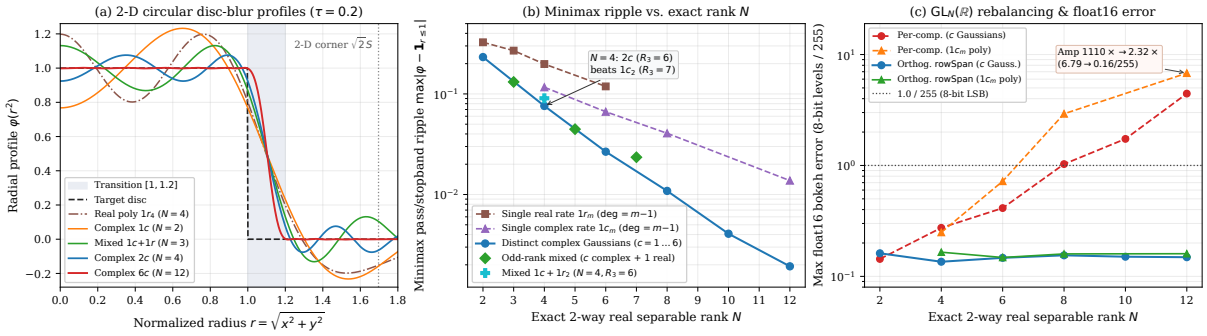


Figure 2: Two-dimensional circular disc blur ($\tau = 0.2$, `Applications/blur_2d.py`): (a) minimax radial profiles $\varphi(r^2)$ against the unit disc; (b) minimax pass/stopband ripple across exact 2-way real separable ranks $N \in [2, 12]$ for distinct complex Gaussians, odd-rank mixed Gaussians, and repeated-rate polynomial-Gaussians; (c) maximum IEEE `float16` bokeh rendering error on a 512×512 HDR scene (peak 8.0, radius $R_{\text{px}} = 25$, tap width $W = 61$) before and after nuclear-norm-optimal $\text{GL}_N(\mathbb{R})$ rebalancing on rowSpan(K).

Table 3: Representative 2-D circular disc-blur designs ($\tau = 0.2$, $R_{\text{px}} = 25$, $W = 61$). Exact 2-D rank N and 3-D ball rank R_3 follow Theorem 3.5; SVD is the empirical singular-value count ($\sigma_k > 10^{-10}\sigma_1$) of the 61×61 kernel; $\kappa_\infty = (\sum_n \|h_n\|_\infty \|v_n\|_\infty)/\|K\|_\infty$ measures peak channel amplification before (raw) and after (orth) rowSpan rebalancing, alongside max `float16` HDR bokeh error (in 8-bit units /255).

Structure	Degrees	N (d=2)	SVD	R_3 (d=3)	Ripple $\pm\delta$	$\kappa_\infty^{\text{raw}}$	$\kappa_\infty^{\text{orth}}$	`fp16` raw	`fp16` orth
$1c$ (Gaussian)	0	2	2	3	0.2325	1.65	1.30	0.14	0.16
$1c + 1r$ (mixed)	0, 0	3	3	4	0.1315	3.45	1.65	0.15	0.15
$2c$ (Gaussians)	0, 0	4	4	6	0.0759	5.28	1.95	0.27	0.14
$1c + 1r_2$ (mixed poly)	0, 1	4	4	6	0.0905	4.50	1.92	0.20	0.15
$1c_2$ (complex poly)	1	4	4	7	0.1165	5.68	1.88	0.25	0.17
$1r_4$ (real Laguerre)	3	4	4	7	0.1982	4.74	1.56	0.31	0.15
$2c + 1r$ (mixed)	0, 0, 0	5	5	7	0.0446	8.73	2.08	0.25	0.13
$4c$ (Gaussians)	0^4	8	8	12	0.0109	70.06	2.31	1.03	0.15
$6c$ (Gaussians)	0^6	12	12	18	0.0019	595.01	2.49	4.44	0.15
$1c_6$ (complex poly)	5	12	12	23	0.0138	1,109.75	2.32	6.79	0.16

1. **Role of nonreal rates, odd budgets, and 3-D lifting:** Across all 18 tested families, empirical SVD rank matches c.multiCost(2) identically. Distinct complex Gaussians achieve geometric ripple decay with $N = 2c$ ($0.2325 \to 0.0019$ from $1c$ to $6c$, or 0.0014 under global Remez search [21]), while adding a single real Gaussian ($1c+1r, 2c+1r$) interpolates cleanly at odd ranks $N = 3, 5$. Free real-only rates coalesce into repeated-rate Laguerre–Gaussians ($1r_m$). At $N = 4$, four distinct structures tie in 2-D cost, but lifting to a 3-D spherical ball splits their exact 3-way ranks: $2c$ gives both the lowest ripple (± 0.0759) and lower 3-D rank ($R_3 = 6$ vs. $R_3 = 7$ for $1c_2$ and $1r_4$).

2. **Transitive $\text{GL}_N(\mathbb{R})$ freedom in $d = 2$ vs. restricted CP orbits in $d \geq 3$:** While $N = \dim \text{rowSpan}(K)$ is an algebraic invariant, in $d = 2$ *every* $T \in \text{GL}_N(\mathbb{R})$ maps $(h, v) \mapsto (Th, T^{-\top}v)$ to an equivalent rank-N realization, allowing arbitrary orthogonal mixing *across* distinct components. Splitting high-order approximations component-by-component causes severe intermediate cancellation ($\kappa_\infty^{\text{raw}} = 595$ on $6c$ and 1,110 on $1c_6$), producing visible 4.44/255 and 6.79/255 errors around HDR specular highlights when intermediate passes are stored in IEEE `float16` (as in WebGPU `pack2x16float` [21]). Orthonormalizing the 1-D basis via QR and symmetric eigendecomposition $K(x, y) = \sum_{n=1}^{N} e_n g_n(x) g_n(y)$ ($\|g_n\|_2 = 1$, $h_n = \sqrt{|e_n|}\, g_n$, $v_n = \text{sign}(e_n) h_n$, accumulated in ascending order of $|e_n|$)

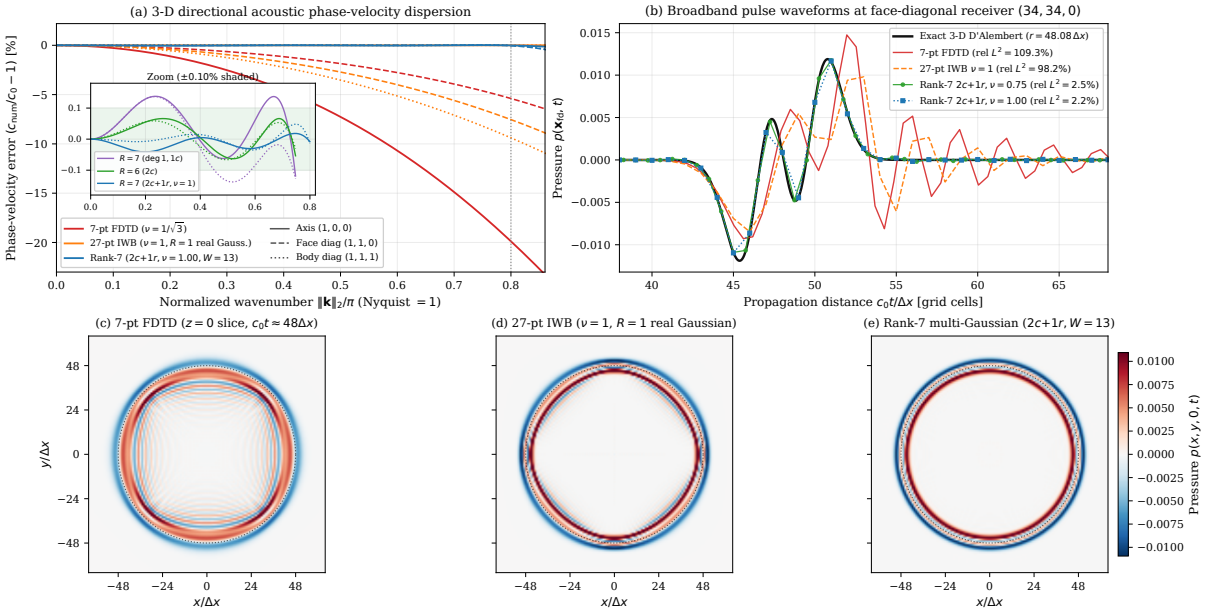


Figure 3: Three-dimensional acoustic wave propagation (`Applications/acoustics_3d.py`): (a) directional phase-velocity error ($c_{\text{num}}(\mathbf{k})/c_0 - 1$) along $(1,0,0)$, $(1,1,0)$, and $(1,1,1)$ across $\|\mathbf{k}\|_2/\pi \in [0, 0.86]$, with inset zoom ($\pm 0.10\%$ shaded) for $R = 6, 7$; (b) broadband pulse waveforms ($\sigma_0 = 1.40\,\Delta x$) on a 256^3 grid at the face-diagonal receiver $(34, 34, 0)$ ($r = 48.08\,\Delta x$) vs. the exact 3-D D'Alembert solution; (c–e) mid-plane wavefield slices $p(x, y, 0, t)$ at $c_0 t \approx 48\,\Delta x$ (dotted circle: $r = 48\,\Delta x$).

achieves the global nuclear-norm minimum $\sum_n \|h_n\|_2 \|v_n\|_2 = \sum_n |e_n| = \|K\|_{S_1}$, collapsing κ_∞ to ≤ 2.49 and keeping `float16` max error at 0.13–0.17/255 (RMSE $\leq 0.033/255$) across all $N \leq 12$. By contrast, in $d \geq 3$, minimal d-way CP decompositions do not admit transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations (under Kruskal's sufficient k-rank condition $k_1 + k_2 + k_3 \geq 2R + 2$ [11] minimal trilinear decompositions are unique up to permutation and scaling, and even where $2 \times 2 \times 2$ complex-Gaussian blocks retain internal gauge parameters, cross-component cancellation cannot be orthogonally rotated away without raising rank), making well-separated rates and amplitude-balanced root-of-unity factors essential.

3. **GPU throughput:** Using two-pass symmetric Triton kernels with constant two-buffer memory on 1920×1080 RGBA images at radius $R_{\text{px}} = 64$ ($W = 155$), direct 155×155 depthwise `conv2d` takes 137.1 ms, whereas rank $N = 4, 8, 12$ takes $7.01, 14.17, 21.19$ ms in `float32` ($19.6\times, 9.7\times, 6.5\times$) and $3.99, 8.05, 12.09$ ms in `float16` ($34.3\times, 17.0\times, 11.3\times$).

5.3 Worked application II: isotropic 3-D acoustic wave simulation

Consider the 3-D constant-speed scalar acoustic wave equation $\partial_{tt} p = c_0^2 \nabla^2 p$ discretized on $\mathbb{Z}^3$ with spacing Δx and time step $\Delta t = \nu \Delta x / c_0$ (Courant number $\nu > 0$) by the explicit two-step leapfrog update

$$p^{n+1}[\mathbf{m}] + p^{n-1}[\mathbf{m}] = (M * p^n)[\mathbf{m}], \qquad \mathbf{m} \in \mathbb{Z}^3. \tag{11}$$

In discrete wavenumber $\mathbf{k} = (k_x, k_y, k_z) \in [-\pi, \pi]^3$, the exact continuous wave-equation propagator has radial symbol $\widehat{M}_{\text{exact}}(\mathbf{k}) = 2\cos(\nu \|\mathbf{k}\|_2)$. Spectral von Neumann stability holds when $\widehat{M}(\mathbf{k}) \in [-2, 2]$ (both roots of $\zeta^2 - \widehat{M}(\mathbf{k})\zeta + 1 = 0$ lie on $|\zeta| = 1$), with uniform temporal boundedness and positive discrete energy on nonzero modes whenever $\widehat{M}(\mathbf{k}) \in (-2, 2)$ strictly for $\mathbf{k} \neq \mathbf{0}$ (avoiding repeated unit-circle roots $\zeta = \pm 1$, while $\widehat{M}(\mathbf{0}) = 2$ represents the spatially constant mode). Standard 7-point FDTD ($\nu \leq 1/\sqrt{3}$) suffers up to 19.88% axial phase-velocity

error by $\|\mathbf{k}\|_2 = 0.80\pi$. Two structural facts explain why radial multi-Gaussians are natural here:

- **Inadequacy of additive axial stencils of any order:** Any purely axial scheme $\widehat{M}_{\text{ax}}(\mathbf{k}) = 2+A(k_x)+A(k_y)+A(k_z)$ ($A(0) = 0$) lacks mixed cross-terms $k_x^2 k_y^2$: even if $A(k) = 2(\cos(\nu k)-1)$ has zero dispersion along the coordinate axes, along $(1,1,1)/\sqrt{3}$ it gives $2+6(\cos(\nu k/\sqrt{3})-1) = 2-\nu^2k^2+\frac{1}{36}\nu^4k^4-\ldots \neq 2\cos(\nu k) = 2-\nu^2k^2+\frac{1}{12}\nu^4k^4-\ldots$, leaving an irreducible $O(\nu^4k^4)$ anisotropy gap while Brillouin-corner stability $\widehat{M}_{\text{ax}}(\pi,\pi,\pi) \geq -2$ forces $A(\pi) \geq -4/3$ ($\nu k_c \leq \arccos(1/3) \approx 0.392\pi$ when A tracks $2(\cos(\nu k)-1)$).
- **Kowalczyk–van Walstijn IWB is a single real Gaussian:** In room acoustics, Kowalczyk and van Walstijn's 27-point Interpolated Wideband (`IWB`) stencil [10] ($a = \frac{1}{4}, b = \frac{1}{16}$, $\nu \leq 1$) has face, edge, and corner weights $\frac{1}{4}\nu^2, \frac{1}{8}\nu^2, \frac{1}{16}\nu^2$ and center weight $2-\frac{7}{2}\nu^2$, which is *identically*

$$M_{\text{IWB}}[\mathbf{m}] = (2-4\nu^2)\delta_{\mathbf{0}}[\mathbf{m}] + \frac{\nu^2}{2}\,2^{-(m_x^2+m_y^2+m_z^2)} \qquad (\mathbf{m} \in \{-1,0,1\}^3),$$

  the ($r = 1, c = 0, L = 1, R = 1$) single-real-Gaussian member of our radial family (plus a zero-pass center tap). Because a single positive real Gaussian cannot match higher radial derivatives of $2\cos(\nu\|\mathbf{k}\|_2)$, `IWB` at $\nu = 1$ is exact on axes but incurs 9.38% diagonal phase-velocity error by 0.80π (Figure 3a,d).

Brillouin-octant Chebyshev optimization and exact Neumann walls. Generalizing `IWB` to wider stencils $\mathbf{m} \in [-L, L]^3$, we set $M[\mathbf{m}] = \alpha_0\delta_{\mathbf{0}}[\mathbf{m}] + \varphi(m_x^2+m_y^2+m_z^2)$, where α_0 fixes $\widehat{M}(\mathbf{0}) = 2$ identically and φ has exact 3-way real rank $R = r+3c$ (Proposition 3.2) or $R = 7$ for $\deg P = 1$ (Proposition 3.4; verified to $\leq 5.6\times10^{-16}$ on $[-L,L]^3$). For fixed rates (searched via differential evolution and Nelder–Mead), minimizing $\max_{0<\|\mathbf{k}\|_2\leq k_c} |\widehat{M}(\mathbf{k}) - 2\cos(\nu\|\mathbf{k}\|_2)|/(2\nu\|\mathbf{k}\|_2\sin(\nu\|\mathbf{k}\|_2))$ subject to exact asymptotic sound speed $\partial_{k_x}^2\widehat{M}(\mathbf{0}) = -2\nu^2$ and strict interior stability $-2+10^{-3} \leq \widehat{M}(\mathbf{k}) \leq 2-\min(10^{-3}, \frac{1}{4}\nu^2\|\mathbf{k}\|_2^2)$ across 232,620 symmetry-inequivalent Brillouin-octant and spherical-shell wavevectors (including the exact sphere $\|\mathbf{k}\|_2 = k_c$) is solved via cutting-plane HiGHS linear programming, followed by multi-start continuous Nelder–Mead maximization of $|\delta c(\mathbf{k})/c_0|$ over $0 < \|\mathbf{k}\|_2 \leq k_c$. Moreover, because every 1-D radial factor $h[m] = h[-m]$ is even, half-sample symmetric index reflection at domain boundaries simultaneously diagonalizes all $3R$ 1-D passes in the 3-D `DCT-II` cosine basis—enforcing exact rigid-wall Neumann boundary conditions ($\partial_n p = 0$) without padding buffers or boundary instabilities. Across 1,000 steps (> 18 wall reflections) in a 64^3 rigid enclosure, discrete leapfrog energy drifted by at most 4.5×10^{-7} (`float32` rounding).

Table 4 and Figure 3 summarize accuracy and GPU performance (*at $W = 17$, cuDNN switches `conv3d` to an FFT algorithm, which the local $R = 9$ Triton cascade still outperforms by 13.7×–19.3×). Both dedicated 1-pass baselines (7-pt FDTD and 27-pt IWB) run at 0.323 ms/step (51.9 GVox/s, 92% of peak GDDR6X memory bandwidth). On the 256^3 broadband spherical pulse test ($\sigma_0 = 1.40\,\Delta x$), 7-pt FDTD and 27-pt IWB break into anisotropic dispersive wave trains (109.3% and 98.2% waveform error), whereas `Rank-7` ($2c+1r$) keeps $|\delta c/c_0| \leq$ 0.046%–0.049% (< 1 cent) up to 0.80π (2.47 points per wavelength at $k_{0.1\%} = 0.81\pi$), achieving 1.6%–2.8% broadband waveform error across all axes and diagonals (3.2% with `float16` intermediates at $\nu = 0.75$, since $\|w\|_1 \leq 1.51$). Replacing naive $(r+4c) = 9$ expansion by Proposition 3.2's exact $R = 7$ formula saves 21.0% wall-clock time (5.59 ms vs. 7.08 ms, 41.2× faster than direct 13^3). Under an analytical $(\Delta x)^{-3}(\Delta t)^{-1} \propto k_{\text{tol}}^{-4}$ resolution-scaling model, matching `Rank-7`'s 0.1%-accurate cutoff (0.81π vs. 0.085π for 27-pt IWB at $\nu = 1$) requires a 9.53× finer grid per axis in 27-pt IWB—865× more voxels (519×–649× more buffer memory accounting for 5 `fp32` or 3 `fp32`+2 `fp16` fields vs. 3 in IWB, exceeding 16 GiB at 256^3 reference

Table 4: Numerically optimized 3-D acoustic propagators vs. FDTD baselines on 256^3 voxels (RTX 4070 Ti Super). Here $|\delta c/c_0|$ is the worst-case phase-velocity error up to k_c (0.80π for baselines, refined by multi-start continuous maximization); $k_{0.1\%}$ and $k_{0.5\%}$ are bisected spherical cutoffs keeping $|\delta c/c_0| \le 0.1\%$ and 0.5% across 153 octant directions; Wave L^2 is the worst-case receiver relative L^2 waveform error at $r \approx 48\,\Delta x$; GPU step times compare direct W^3 `conv3d`, naive $(r+4c)$/9-term Triton passes, and exact R-term Triton passes in `fp32` and `fp16`-intermediate precision.

Scheme	ν	W	R (nv)	$\lvert\delta c/c_0\rvert$	$k_{0.1\%}$	$k_{0.5\%}$	Wave L^2	Direct	Naive	Exact `fp32`	Exact `fp16`
7-pt FDTD	0.58	3	—	19.88%	0.060π	0.135π	109.3%	—	—	0.323 ms	—
13-pt $O(\Delta x^4, \Delta t^2)$	0.50	5	—	8.74%	0.099π	0.222π	—	—	—	—	—
27-pt IWB ($1r$)	1.00	3	1 (1)	9.38%	0.085π	0.191π	98.2%	—	—	0.323 ms	—
Rank-4 ($1c{+}1r, 0.75\pi$)	0.75	9	4 (5)	0.293%	0.14π	0.76π	11.6%	76.9 ms	4.05 ms	3.30 ms	2.39 ms
Rank-6 ($2c, 0.75\pi$)	0.75	11	6 (8)	0.066%	0.76π	0.80π	5.2%	139.0 ms	6.30 ms	4.80 ms	3.44 ms
Rank-7 (deg 1 $1c, 0.75\pi$)	0.75	13	7 (9)	0.138%	0.16π	0.78π	7.1%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	0.75	13	7 (9)	0.046%	0.81π	0.85π	2.8%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-7 ($2c{+}1r, 0.80\pi$)	1.00	13	7 (9)	0.049%	0.81π	0.87π	2.5%	230.4 ms	7.08 ms	5.59 ms	4.03 ms
Rank-9 ($3c, 0.85\pi$)	0.75	17	9 (12)	0.044%	0.86π	0.89π	2.0%	104.6* ms	10.02 ms	7.66 ms	5.43 ms

scale) and 8,247× more voxel-steps, corresponding to an estimated 476× speedup in `float32` (660× in `fp16`-intermediate; 20×–28× faster and 94× fewer voxels at a 0.5% tolerance).

5.4 Scope, limitations, and numerical conditioning

Three practical boundaries separate the exact continuous-domain theorems of Sections 2–3 from finite-precision filter design:

1. **Exact finite rank vs. uniform approximation:** Theorems 3.3–3.5 characterize *exact* representations of *continuous* radial kernels on $\mathbb{R}^d$. Smooth non-exponential-polynomial profiles (such as $e^{-\|x\|^4}$ or $\omega_c J_1(\omega_c r)/(2\pi r)$) have infinite separable rank by Theorem 3.3; discontinuous profiles require separate analysis (e.g. $\mathbf{1}_{\{\|x\|=0\}} = \prod_j \mathbf{1}_{\{x_j=0\}}$ has rank 1, whereas $\mathbf{1}_{r\le 1}$ has infinite rank via triangular submatrices). While finite sums of real Gaussians $\sum_{m=1}^{M} a_m e^{-m\|x\|^2}$ are uniformly dense in $C_0([0,\infty))$ by Stone–Weierstrass (since $s \mapsto e^{-s}$ separates points and vanishes at ∞), density alone gives neither convergence rates nor bounded weights; achieving rapid Chebyshev decay without extreme cancellation requires transition bands and nonreal oscillatory rates (Figure 2b).

2. **Conditioning of root-of-unity identities at high dimension d or degree k:** In Propositions 3.2 and 3.4, the prefactor $\frac{2^{d-1}}{R}$ sums R alternating products, so unit-modulus complex factors satisfy $\sum_{r=0}^{R-1} \frac{2^{d-1}}{R} \prod_j |\operatorname{Re}(\cdots)| \le 2^{d-1}$. For $d = 2, 3, 4$ ($2^{d-1} = 2, 4, 8$) this amplification is tiny, whereas for large $d \gg 1$ the 2^{d-1} factor grows exponentially and can amplify rounding error. Similarly, for polynomial degree $k = \deg P \ge 1$ on bounded support $s_j = x_j^2 \in [0, S^2]$, evaluating $U_j(X)$ on the unscaled unit circle $|X| = 1$ mixes $O(1)$ and $O(S^{2k}/k!)$ coefficients; rescaling $X \mapsto \rho X$ with $\rho \approx S^{-2}$ (replacing $[X^k]\prod_j U_j(X)$ by $\rho^{-k}[X^k]\prod_j U_j(\rho X)$) balances the monomial terms across the evaluation circle, while balanced scalar roots ($|a|^{1/d}$) equalize factor norms. At high multiplicity $k \gg 1$ or near-coalescing rates (compare Theorem 3.7's $O(h^{-1})$ weights), $d = 2$ permits exact *a posteriori* $\mathrm{GL}_N(\mathbb{R})$ orthogonalization across components on $\mathrm{rowSpan}(K)$ (Figure 2c), whereas the absence of transitive $\mathrm{GL}_R(\mathbb{R})$ cross-component rotations in $d \ge 3$ favors distinct complex Gaussians ($k = 0$) over high-degree repeated rates.

3. **Discretization, truncation, and heuristic rate search:** The Lean formalization treats continuous functions on $\mathbb{R}^d$; grid sampling on $\mathbb{Z}^d$ (which can lower discrete rank at aliased frequencies $\operatorname{Im}\lambda \in \pi\mathbb{Z}$), finite box truncation $[-S, S]^d$, heuristic nonconvex rate search

Table 5: Structure and source line counts of the `KernelRankLean` library.

Module(s)	Mathematical content	Phys.	Nonbl.
`Defs`, `KernelRankLean`	Multi-way rank, surjections, slices, costs	242	207
`RowSpan`, `Complexification`	2-way row-span rank; $\mathbb{R} \to \mathbb{C}$ invariance	221	189
`Translations`	d-D radial-to-Hankel reduction; `translateSpan`	92	80
`ExpPoly`, `RealExpPoly`	Exponential-polynomial uniqueness	198	170
`Semigroup`, `TranslationSemigroup`	Banach semigroups; shift semigroup	262	219
`Necessity`, `Representation`	Generalized eigenvectors; conjugate pairing	222	198
`Differential`, `ComplexRank`	Half-line operators $q(D)$; 2-D complex rank	318	284
`Annihilator`	Minimal constant-coefficient annihilator	69	62
`MultiComplexRank`	Exact d-way complex rank $\sum_i((d-1)\deg P_i + 1)$	1,015	978
`MultiRealRank`	Exact d-way real rank (`multiTotalCost`)	2,562	2,521
`Main`	Unified d-D/2-D theorems & border-rank gap	377	359
Total (`KernelRankLean`)		**5,578**	**5,267**

(differential evolution plus Nelder–Mead around exact inner Chebyshev linear programs), and IEEE `float16`/`float32` rounding lie outside the formal theorems and are evaluated empirically in Sections 5.2–5.3.

6 Formalization Architecture and Verification Audit

Table 5 breaks down the `KernelRankLean` library (5,578 physical lines, 5,267 nonblank lines; earlier stages comprised 5,324 physical lines at commit `9722ff5` prior to Theorem 3.7, 3,822 lines at commit `6aabb4d` for d-D complex and multi-Gaussian ranks, 1,587 lines at commit `cae13d6` for 2-D plus differential annihilators, and 1,340 lines at commit `240dea0` for 2-D classification alone). Three formalization choices proved decisive in Lean 4:

1. **Dependent multi-way definitions with homogeneous two-coordinate specialization:** Defining `IsSumOfProducts` and `separableRank` over a general dependent family $\prod_{i\in\iota} X_i$ unifies 2-D and d-D statements, whereas specializing `rowSpan` in `RowSpan.lean` to homogeneous $\mathrm{Fin}\,2 \to \alpha$ (with a pattern-matched constructor `fin2Vec`) avoids expensive higher-order unification timeouts when rewriting on $\mathrm{Fin}\,2 \to \mathbb{R}_{\geq 0}$.
2. **Explicit uniformity selection on `translateSpan`:** Transferring a norm from $\mathbb{C}^n$ onto the finite-dimensional subspace $\mathcal{W}(\psi) \leq (\mathbb{R}_{\geq 0} \to \mathbb{C})$ creates a second `UniformSpace` instance alongside the subtype uniformity; explicitly installing the norm-induced uniformity as the local instance prevents typeclass diamond timeouts while letting `LinearMap.continuous_of_finiteDimensional` supply automatic continuity.
3. **Cayley–Hamilton divisibility in place of cyclic vectors:** Bounding $\dim_{\mathbb{C}} \mathcal{W}(\psi)$ from below via divisibility of $\chi_A(X)$ by $\prod_\mu (X-\mu)^{\deg P_\mu + 1}$ eliminated both coefficient-synthesis operators and cyclic-vector constructions, shrinking the 2-D proof plus differential-annihilator extension by 93 lines while yielding Theorem 3.6 directly.

Verification audit. A single script (`scripts/check`, also wired as the repository's pre-commit hook) builds both `KernelRankLean` and `Experiments` via `lake build` (which includes every `.lean` module in the repository), checks the compiler log to ensure no declaration uses `sorry`, runs `#print axioms` on 20 top-level theorems to confirm dependence solely on Lean's three standard classical axioms (`propext`, `Classical.choice`, `Quot.sound`), builds the Isabelle2026 session `KernelRank_NSA`, and audits `Semigroup_NSA.semigroup_eq_exp` via `Thm_Deps.all_oracles` (catching `skip_proof`/`sorry` even under `quick_and_dirty`) while rejecting any `axiomatization` or `axioms` declarations.

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This work builds directly on problems, mathematical formulations, and software developed by Olli Niemitalo, including `kernel-optimizer` [21] for complex-Gaussian approximations of 2-D circularly symmetric blur kernels and his investigations of isotropic low-pass filters, steerable gradient orientation estimation, and planar harmonic differential operators [17, 18, 19, 20]. Olli Niemitalo initiated and guided the `kernel-rank-lean` repository, proposed comparing standard and Isabelle/HOL nonstandard proofs of the semigroup step, and suggested extending the formalization to d dimensions as well as adding the visual pipeline exposition and 2-D/3-D numerical applications. We also thank the aiXiv Official Agent reviewers on versions 1.0 and 1.1, whose constructive critiques motivated Theorems 3.5 and 3.7, Proposition 3.4, the notation glossary, and the conditioning analysis. Both the repository and this manuscript are released under CC0 1.0 Universal.

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